

OXFORD

macroeconomics

INSTITUTIONS, INSTABILITY, AND INEQUALITY

WENDY CARLIN • DAVID SOSKICE

Macroeconomics for International Commerce

PART 1. Flipped Classroom – Week 3 Pre-Class Video Slides

Chapter 3: The 3-Equation Model and Macroeconomic Policy

How to use this video

Before class:

- Watch once straight through
- Try to catch the intuition, stop and think
- Bring one question to class
- In class: we will talk about more technical side and applications

Purpose of this video: *learn the intuition only. Technical derivations and diagrams will be done in class.*

What we cover today

- Why the central bank cares about inflation.
- Why inflation targeting is used.
- The three building blocks: IS, PC, and MR.
- IS curve: demand side, interest rate affects output.
- PC curve: supply side, output gap affects inflation.
- MR / policy rule: central bank responds to inflation and output gaps.
- One simple demand-shock example.
- Why policy involves trade-offs and lags.

Question for today

How does the central bank react to a demand shock?

Why the central bank cares about inflation

- Inflation is not just a change in one price. It is a change in the general price level.

High / volatile inflation

- harder planning for firms
- less reliable contracts
- distorted relative prices

Deflation

- real debt burden rises
- spending delayed
- output can fall further

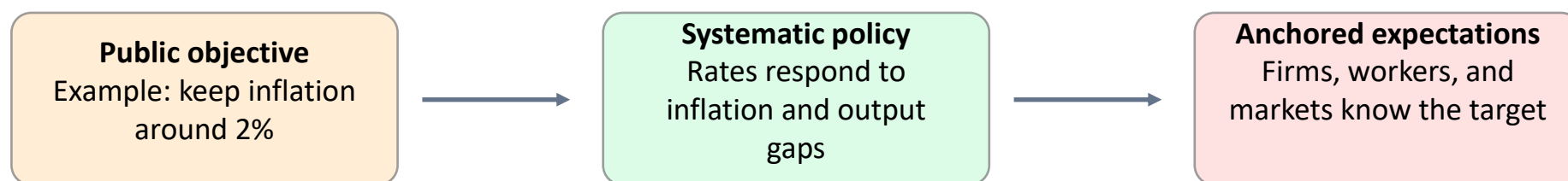
Very low inflation target?

Central banks usually prefer low but positive inflation.

Key idea: the central bank tries to stabilize inflation without creating unnecessary output loss.

Why inflation targeting is used

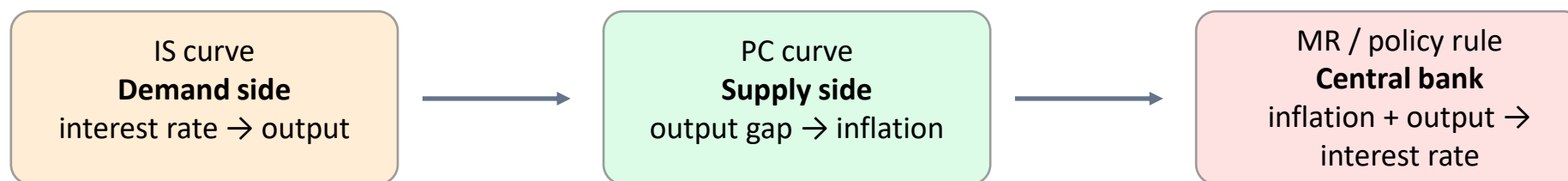
- Inflation targeting gives monetary policy a clear anchor.



- Why delegate to an independent central bank?
 - Monetary policy can often react faster than fiscal policy.
 - Credibility helps prevent inflation expectations from drifting.
 - But monetary policy cannot directly fix supply-side unemployment.

The three building blocks: IS, PC, and MR

- The 3-equation model combines demand, supply, and policy.



- In words: *A shock changes demand and output. The output gap changes inflation. The central bank changes the interest rate to move output and inflation back toward target.*

IS curve: demand side (from Week 1)

- The IS curve shows how the real interest rate affects aggregate demand and output.

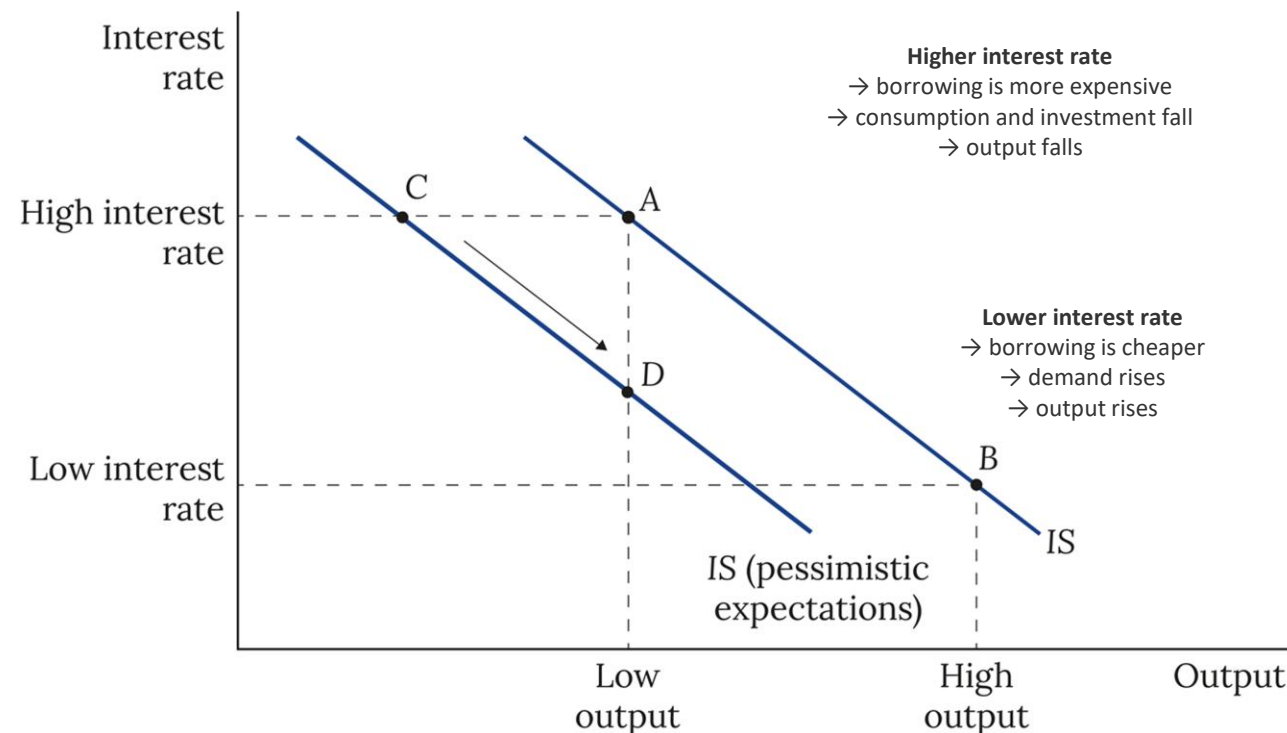
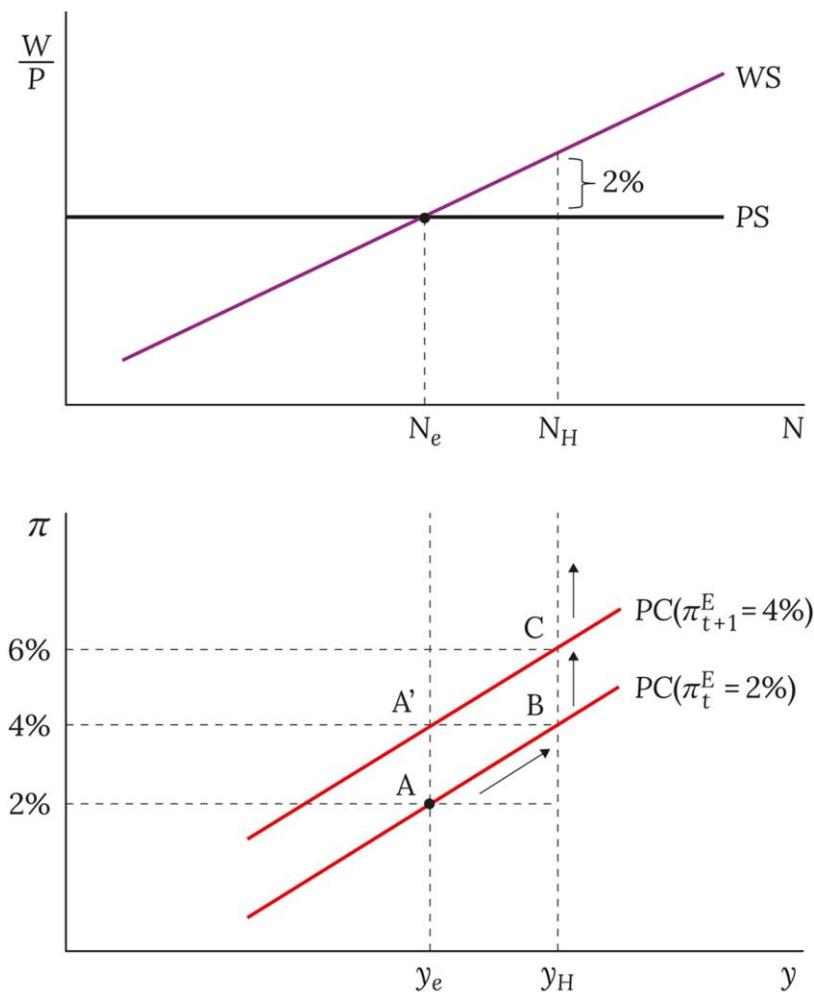


Figure 1.4 The IS curve---the effects of changes in optimism and economic policy. Visit www.oup.com/he/carlin-soskice to explore the IS curve with Animated Analytical [Diagram 1.4](#).

We model the demand side with Investment-Savings Curve (IS Curve)

- Downward Sloping
(Higher interest rate associated with lower AD)
- Examples of factors that shift IS
 - Expectations of the future
(pessimistic expectations → lower AD at every interest rate)
 - Value of collateral in housing
(value of collateral reduces → lower AD at every interest rate)

PC curve: supply side (from Week 2)



The Phillips Curve connects the output gap to inflation pressure.

Output above potential

- unemployment low
- wage pressure rises
- inflation rises

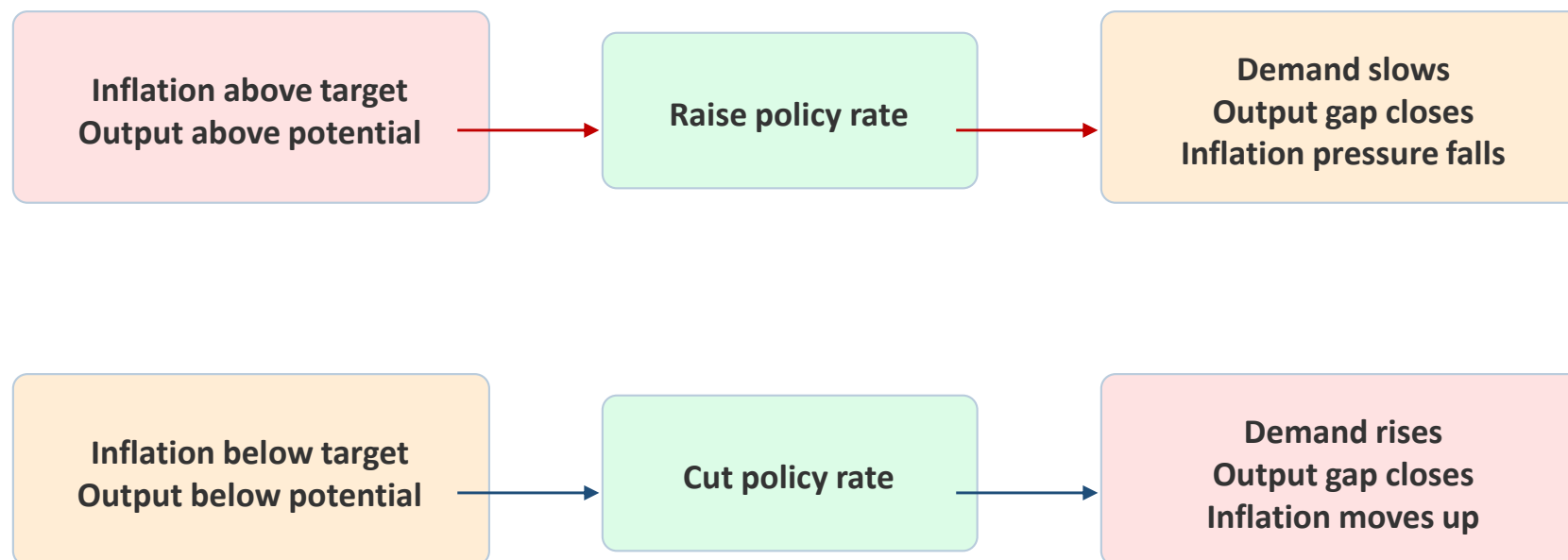
Output below potential

- unemployment high
- wage pressure weakens
- inflation falls

Figure 2.17 The derivation of the Phillips curve. Visit www.oup.com/he/carlin-soskice to explore the derivation of the Phillips curve with Animated Analytical Diagram 2.17.

MR / policy rule: central bank response

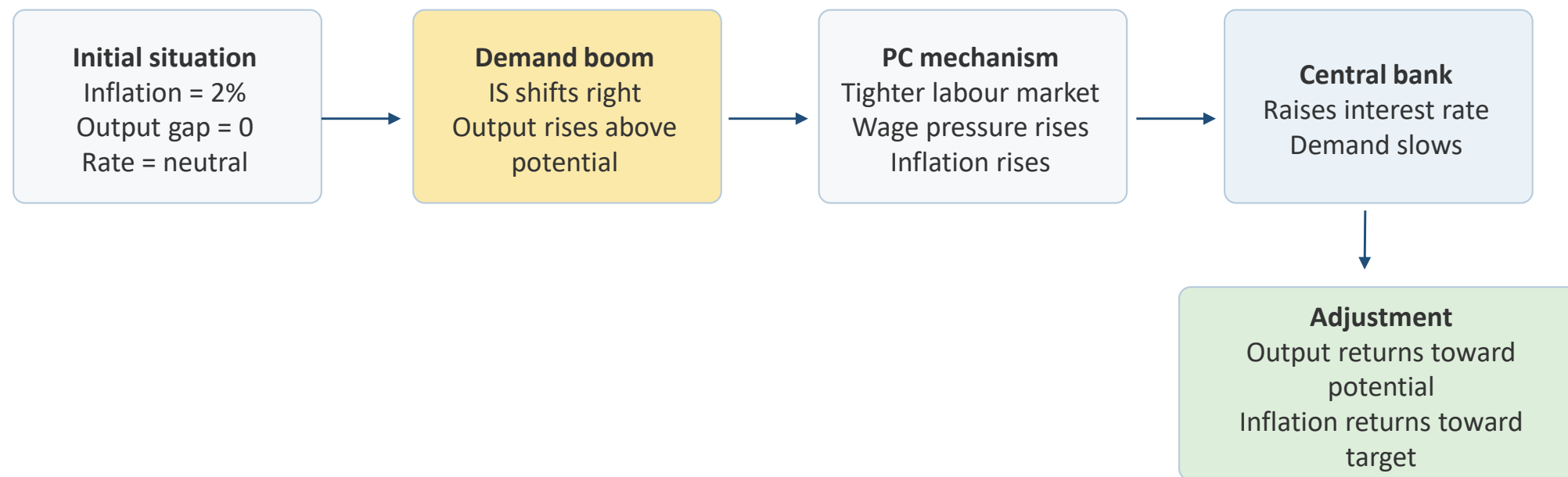
- The central bank uses the interest rate to move the economy back toward target.



This is the intuition behind the MR curve and Taylor-rule style thinking.

Simple example: positive demand shock

- A global demand boom increases spending on domestic goods.



Main point: the shock begins on the IS side, creates inflation pressure through the PC, and triggers the central bank response through MR.

Why policy involves trade-offs and lags

- Central banks cannot perfectly stabilize everything immediately.

Trade-off

Reducing inflation may require lower output for some time.

Lags

Interest-rate changes affect spending, output, and inflation with delay.

Forecasting

The central bank must judge whether a shock is temporary or persistent.

For international commerce, this matters because policy decisions affect exchange rates, borrowing costs, demand, investment, and country risk.

Readiness check

After watching, you should be able to answer these before class.

1. Why do central banks care about high and unstable inflation?
2. In the 3-equation model, which curve represents the demand side of the economy?
3. What does the Phillips Curve explain in this model?
4. Suppose output is above potential output and inflation is rising above target. According to the basic policy-rule logic, what should the central bank normally do?
5. A positive demand shock increases spending in the economy. What is the likely first effect?
6. Why does monetary policy involve lags?
7. Why might a central bank face a trade-off after a supply shock, such as a sudden rise in energy prices?

For class, remember the causal chain: **shock** → **output gap** → **inflation pressure** → **interest-rate response**.

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PART 2. Full In-Class Slides

Chapter 3: The 3-Equation Model and Macroeconomic Policy

Objectives

By the end of this chapter, students should understand the following:

- The role of the central bank in inflation targeting
- The Monetary Rule (*MR*) curve
- The 3-equation model (*IS-PC-MR*)
- How inflation is stabilized in the 3-equation model
- The deflation trap and the 3-equation model
- Supply shock and the 3-equation model

Overview: inflation-targeting monetary policy regime

Why define the Central Bank's role to be mainly that of controlling inflation?

- The equilibrium rate of unemployment at which inflation is constant is determined by the supply-side (Ch. 2)
- As the Central Bank lacks supply-side policy instruments, it is not the right policy maker to reduce equilibrium unemployment

Why is inflation a concern to governments?

- One reason is electoral motivation as the level of inflation is a key priority for voters

	1 Fully agree	2 Somewhat agree	3 Undecided	4 Somewhat disagree	5 Completely disagree	
US all	52%	32%	4%	8%	4%	n = 117
Economists	18%	28%	11%	26%	18%	n = 80

Table 3.1 Public attitudes on the importance of preventing high inflation. Do you agree that preventing high inflation is an important national priority, as important as preventing drug abuse or deterioration in the quality of our schools?

Source: Survey data from Shiller (1997).

Overview: inflation-targeting monetary policy regime

Why do governments hand over inflation targeting to the CB?

- Monetary Policy is a preferred stabilization tool
 - it is considered politically-neutral and more efficient, as implementing Fiscal Policy require lengthy parliamentary processes
- Governments lack low inflation credibility:
 - independent central banks are free from political pressures and hence have higher credibility in delivering low and stable inflation

Overview: the role of the Central Bank

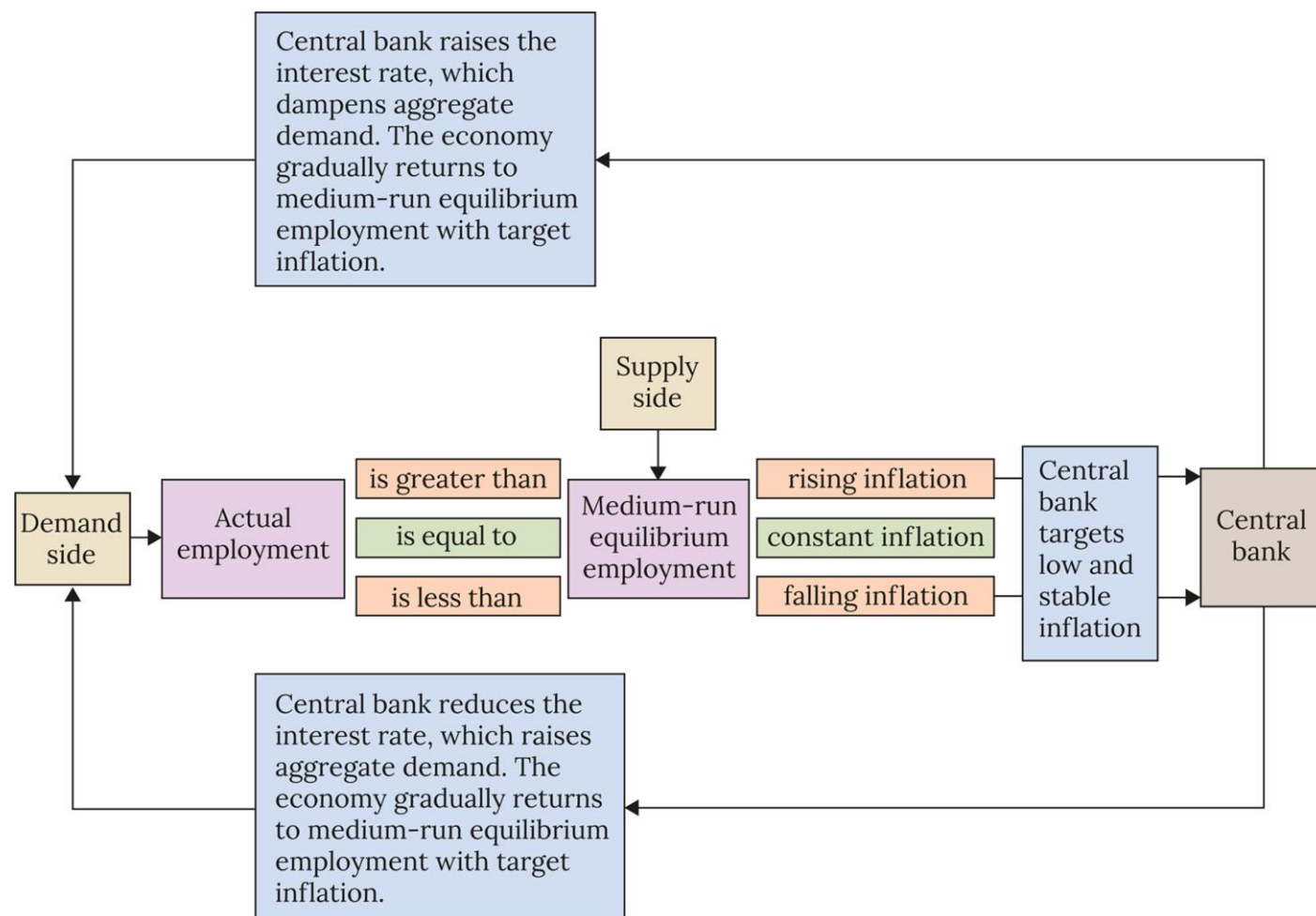


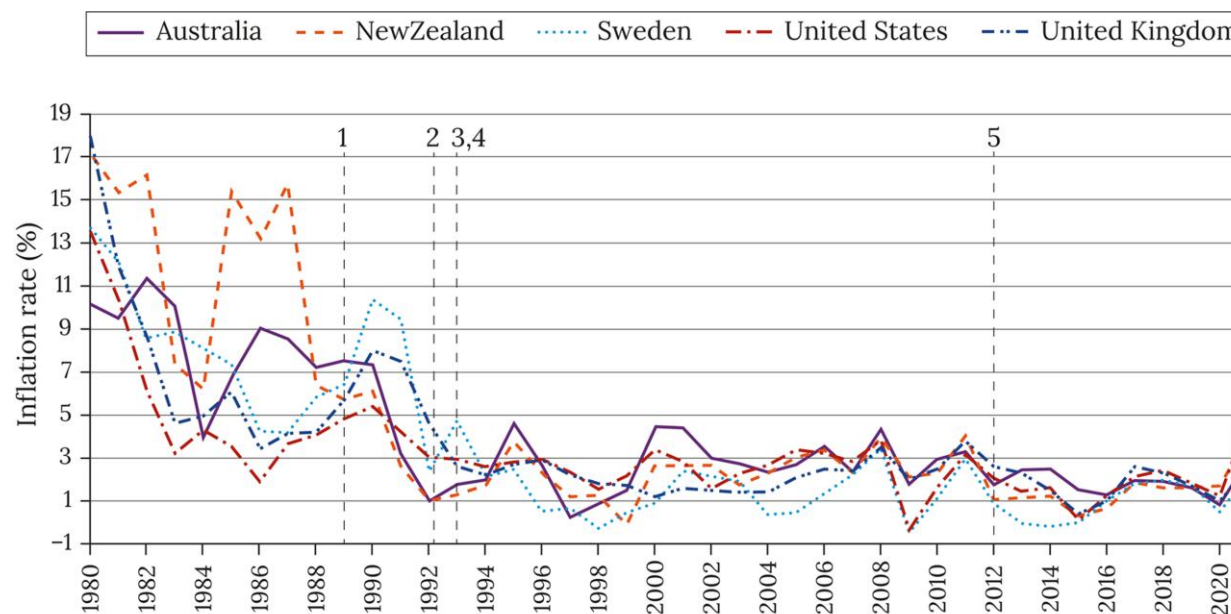
Figure 3.1 Schematic overview of the macro model.

Demand stabilization by changing the interest rate

- Two cases where shocks cannot be stabilized through adjusting interest rate
 - Zero lower bound
 - The economy has a fixed exchange rate

Overview: history of adopting inflation targeting

The adoption of inflation targeting is associated with lower and more stable inflation



Dates inflation targeting was first adopted:

1. Reserve Bank of New Zealand—April 1988
2. Bank of England—September 1992
3. Sveriges Riksbank (the Swedish central bank)—January 1993
4. Reserve Bank of Australia—March 1993
5. US Federal Reserve—January 2012

Figure 3.2 Inflation rates—before and after the adoption of inflation targeting: 1980–2021.

Source: OECD CPI complete database (accessed July 2022).

Overview: inflation and distributional conflict

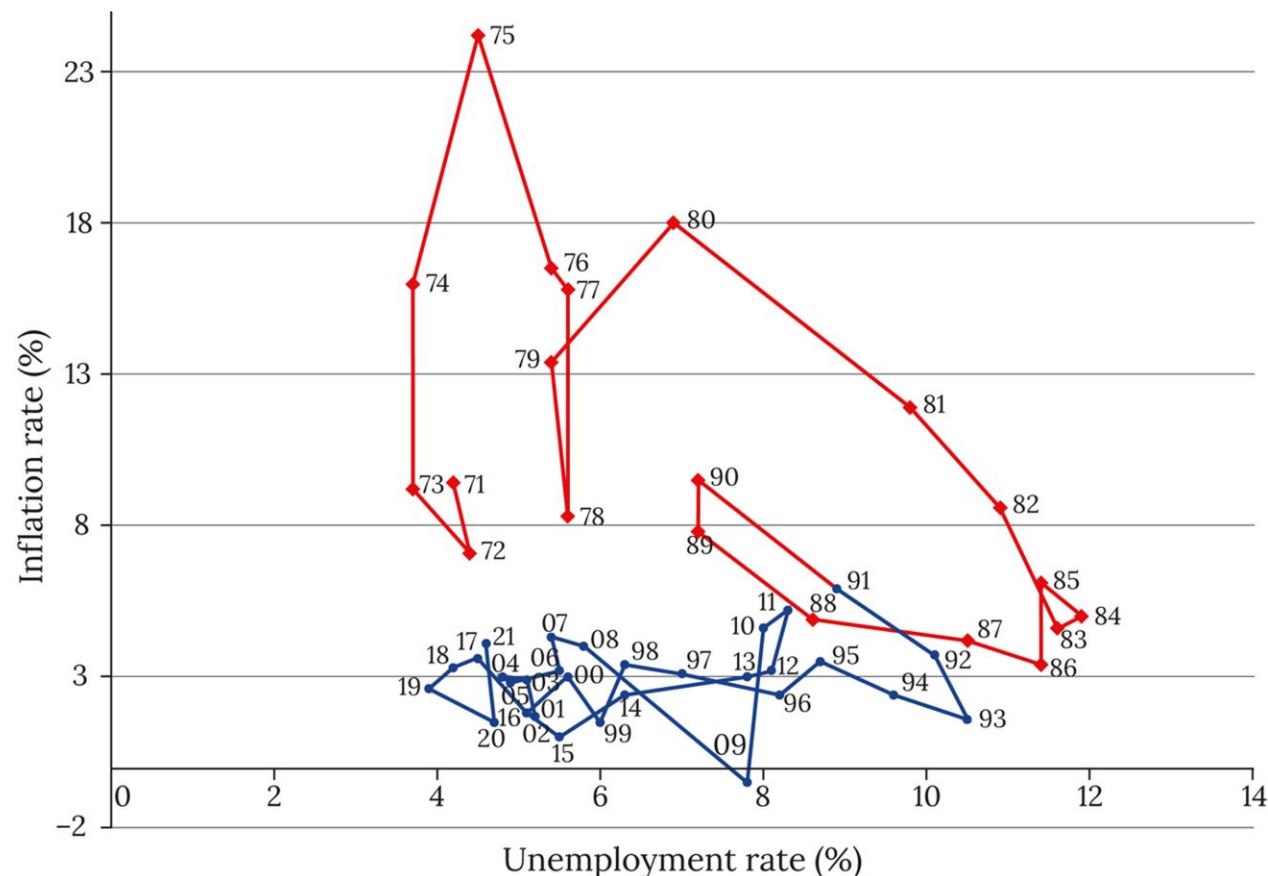


Figure 3.3 UK inflation and unemployment: 1971–2021. The Bank of England adopted inflation targeting in 1992 and was made independent in 1997.

Source: UK Office for National Statistics (data accessed June 2022).

Rising inflation reflects a distributional conflict between owners and employees:

- Lower unemployment → ↑ employees' bargaining power → ↑ real wage → ↑ prices as firms secure their profit margin → inflation
- Rising inflation is usually followed by painful periods of disinflation
- UK in the 1980s: disinflationary process involved a period of high unemployment
 - The costs of high unemployment: include waste of resources, erosion of skills.

Overview: benefits of low and stable inflation

Cost of high and volatile inflation

- The public and economists hold different opinions

	1	2	3	
US all	7%	77%	15%	n = 110
Economists	49%	12%	40%	n = 78

Table 3.2 What is people's biggest gripe with inflation? Which of the following comes closest to your biggest grip about inflation:

1. Inflation causes a lot of inconveniences: I find it harder to comparison shop, I feel I have to avoid holding too much cash, etc.

2. Inflation hurts my real buying power, it makes me poorer.

3. Other.

Source: Survey data from Shiller (1997).

- Volatile inflation can distort resource allocation as it blurs any changes in relative prices, thus leading to inefficient allocations
- High inflation can distort behavior of the public:
 - Shoe-leather costs and menu costs

Overview: dangers of deflation

Central Banks target a low but positive inflation rate instead of 0% due to the dangers of deflation:

- The deflation trap
 - Continued weak demand → negative inflation → $\uparrow$ real interest rate
- Increase real debt burdens
 - Deflation increases debt as a proportion of income
- Resistance to nominal wage cuts
 - Inflation makes it easier to reduce real wages when necessary

Overview: the 3-equation model

The 3-equation model: IS, PC and MR Curves

- The IS Curve (Ch. 1) reflects the demand side; The PC Curve (Ch. 2) reflects the supply side
- The Monetary Rule (MR) Curve shows the optimal way in which Central Banks respond to shocks, given their objectives
- e.g., a consumption boom increases inflation above its 2% target → the central bank increases the interest rate to cut inflation. In the process of doing so, it follows the given MR curve

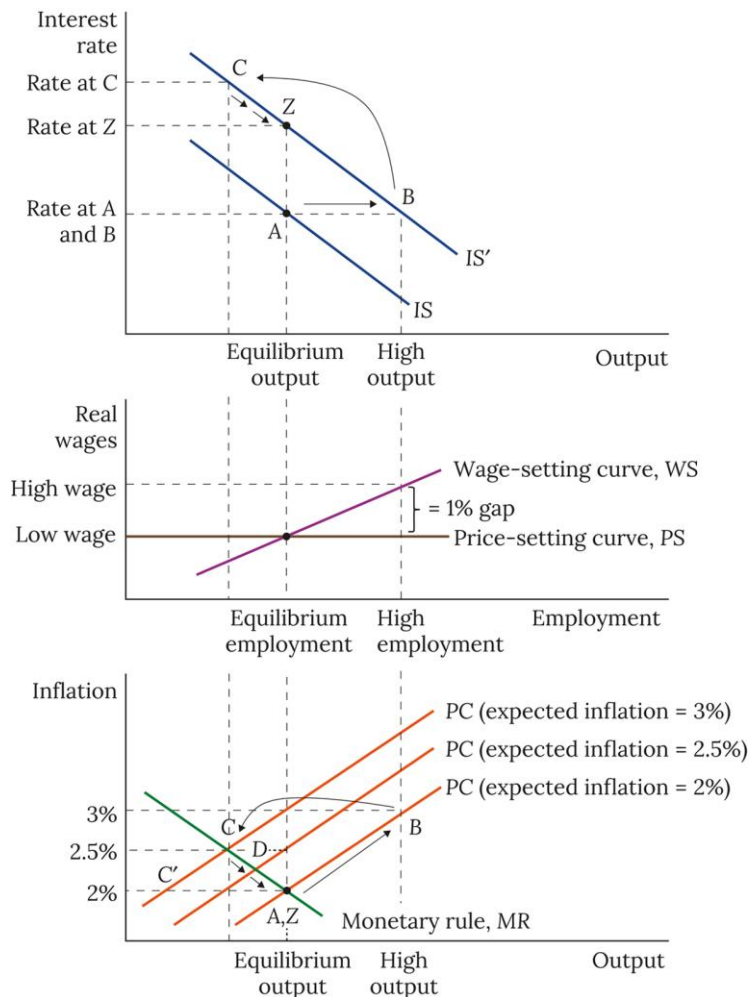
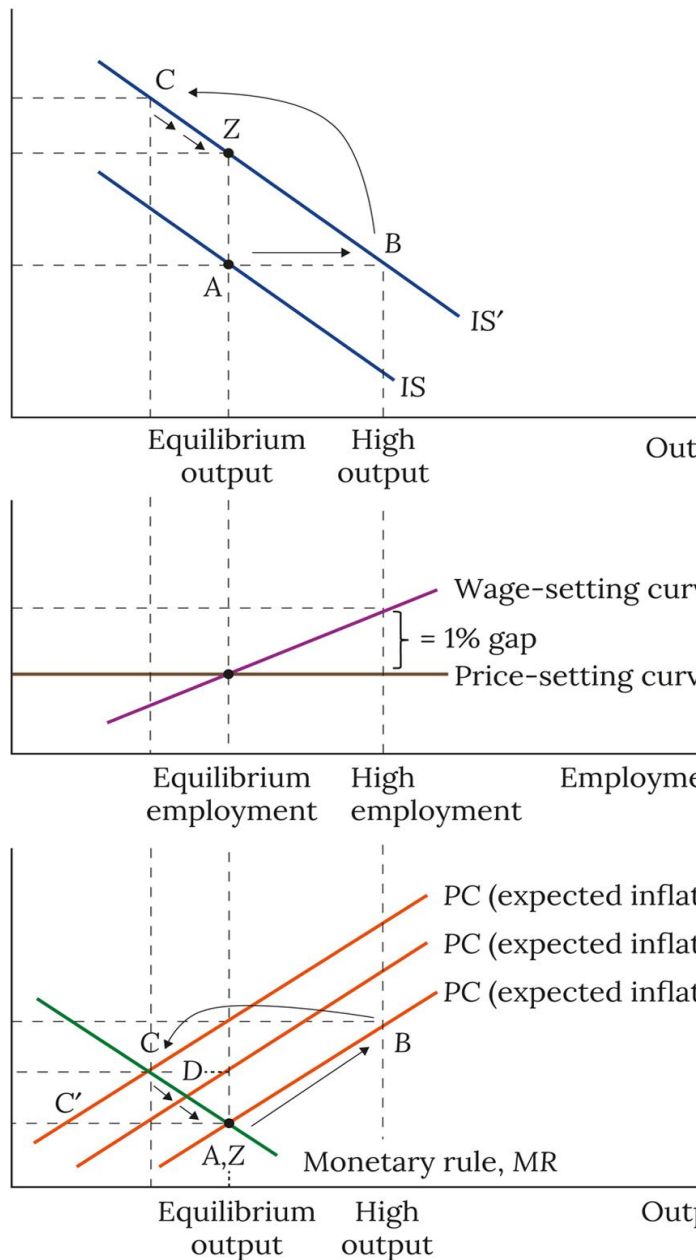


Figure 3.4 The 3-equation model: the adjustment of the economy to a permanent demand shock.

The 3-Equation Model in Action: *Permanent Consumption Boom*



- **Three equations:**

- **IS curve:** $y_t = 0.5 - 0.5r_t + Shock_t$
- **Phillips Curve:** $\pi_{t+1} = \pi_{t+1}^e + 0.5y_t$
- **Taylor Rule:** $i_t = r^* + \pi_t + 1.5(\pi_t - \pi^*) + y_t$

- **Constants:**

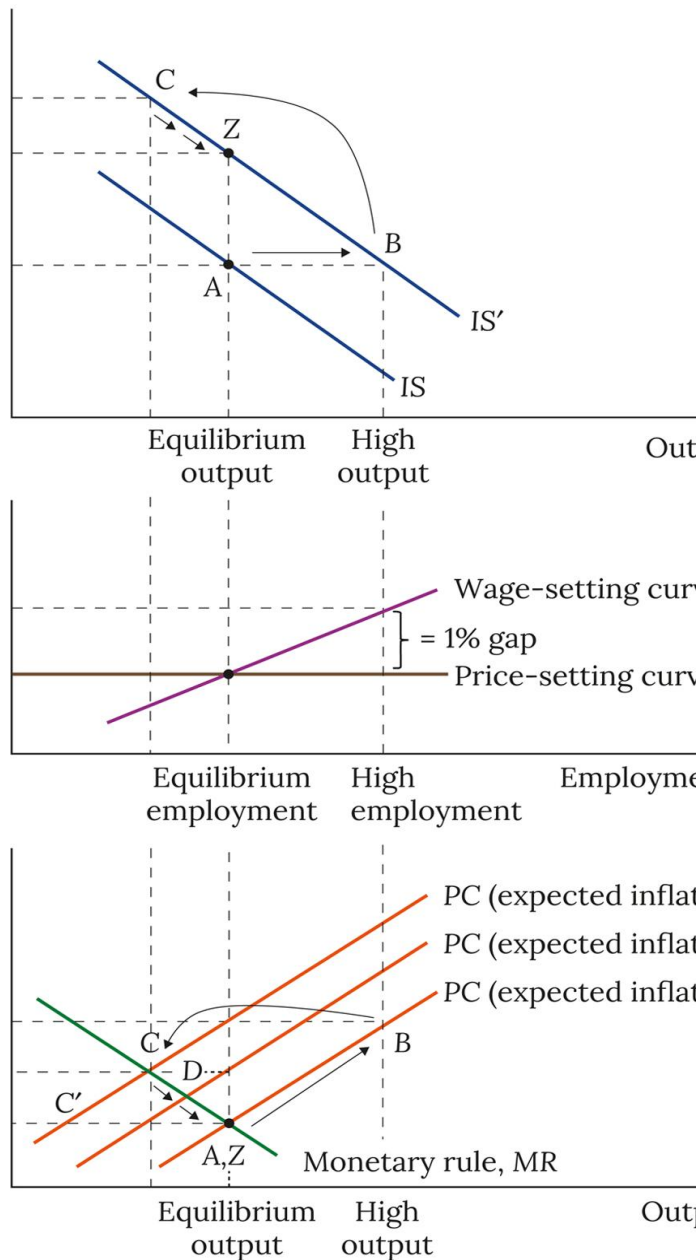
- Target inflation $\pi^* = 2\%$
- Neutral real rate $r^* = 1\%$
- Shock = **+1% permanent consumption boom**

- **Baseline: Point A**

- Inflation: $\pi_0 = 2\%$
- Output gap: $y_0 = 0\%$
- Taylor Rule $\rightarrow i_0 = 3\%$
- Real rate: $r_0 = 1\%$
- IS gives $y_0 = 0$
- **Point A** = equilibrium (target inflation, potential output, neutral rate).

Figure 3.4 The 3-equation model: the adjustment of the economy to a permanent demand shock.

The 3-Equation Model in Action: *Permanent Consumption Boom*

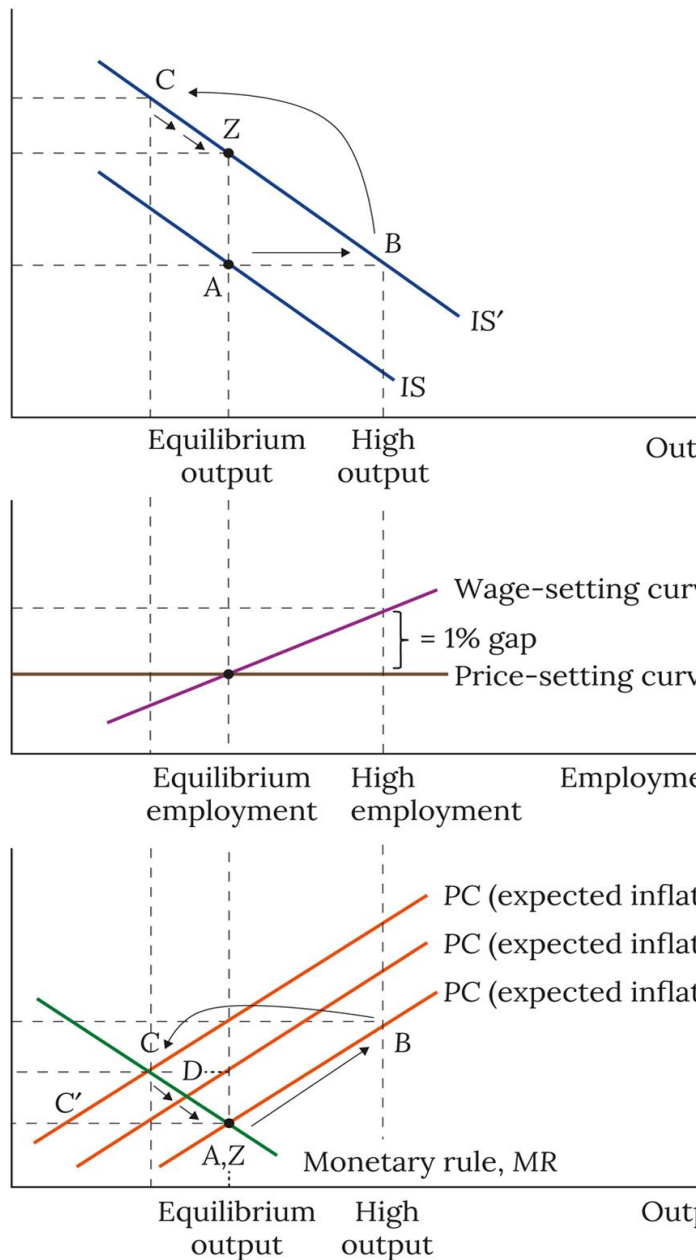


• Shock Hits: Point B

- IS shifts right by +1 (permanent boom)
- Keep old rate: $i = 3\%$
- Real rate: $r = 1\%$
- $IS \rightarrow y_1 = +1\%$
- Labour market tighter $\rightarrow$ bargaining gap = +1%
- Wages rise 3% (2% past inflation + 1% bargaining gap)
- Prices rise 3% $\rightarrow$ Inflation jumps to $\pi_1 = 3\%$
- **Point B** = boom: output +1%, inflation +1pp.

Figure 3.4 The 3-equation model: the adjustment of the economy to a permanent demand shock.

The 3-Equation Model in Action: *Permanent Consumption Boom*

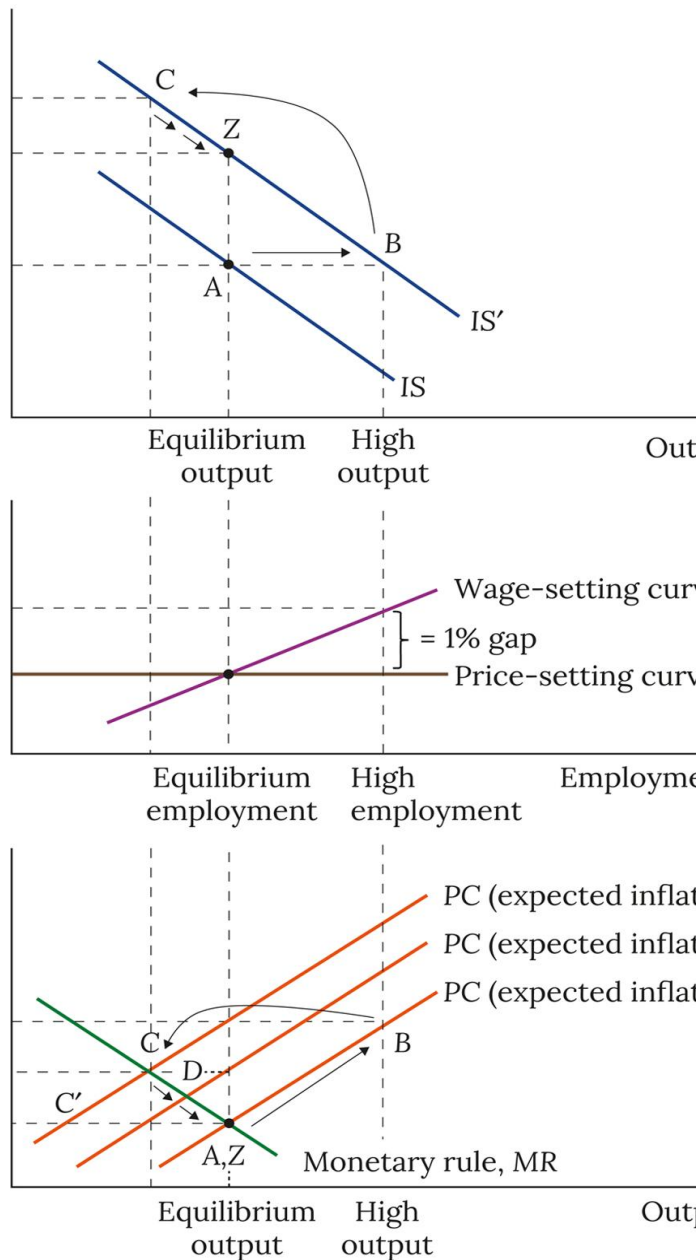


• Central Bank Reacts: Point C

- Taylor Rule with $\pi_1 = 3\%$, $y_1 = +1\%$:
 $i_1 = 6.5\%$
- Real rate: $r_1 = 3.5\%$
- $IS \rightarrow y_1 = +0.75\%$ (boom reduced but not gone)
- Inflation still 3% this period
- **Point C** = policy rate peak, CB follows MR curve.

Figure 3.4 The 3-equation model: the adjustment of the economy to a permanent demand shock.

The 3-Equation Model in Action: *Permanent Consumption Boom*



- **Expectations Adjust: PC Shifts Down**

- Decisive tightening lowers inflation expectations:

$$\pi_2^e = 2.5\%$$

- Phillips Curve now lower:

$$\pi_2 = \pi_2^e + 0.5y_1 = 2.5 + 0.375 = 2.88\%$$

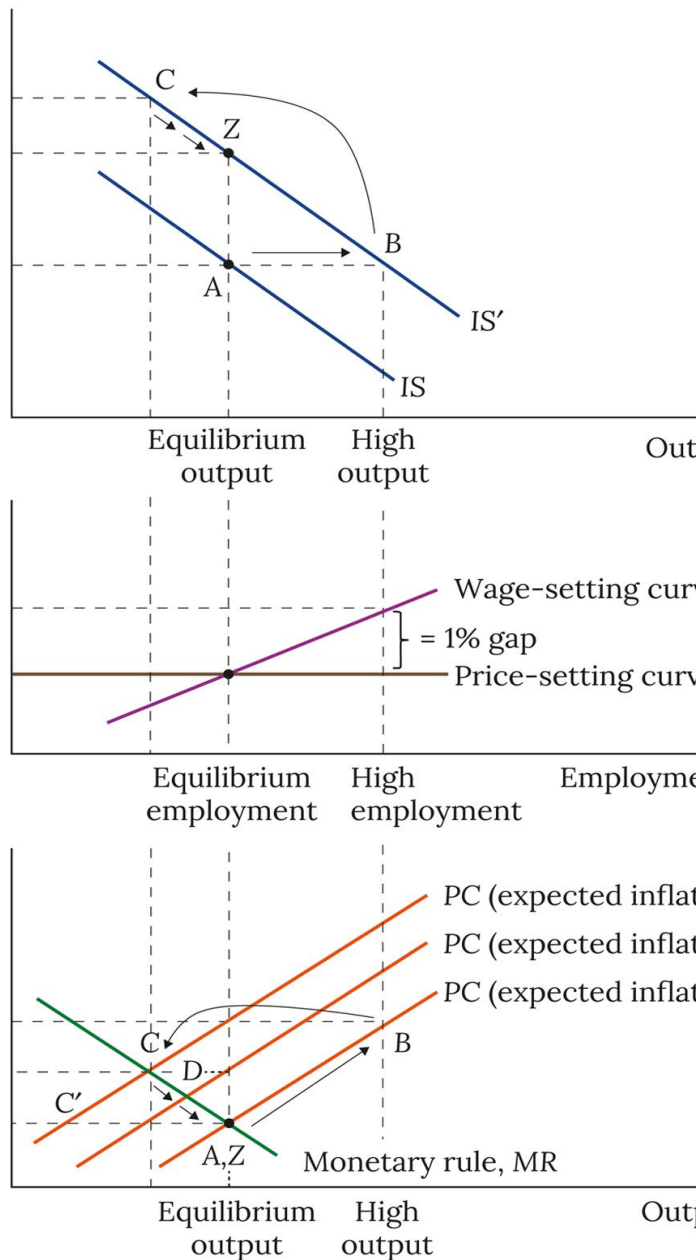
- Inflation easing as PC shifts down.

- **Central Bank Re-optimizes: Point D**

- Use Taylor Rule with $\pi_2 = 2.88\%$:
Solve jointly with $IS \rightarrow y_2 \approx +0.23\%$, $i_2 \approx 5.42\%$, $r_2 \approx 2.54\%$.
- Inflation = 2.88% (falling).
- Policy rate comes down from 6.5% to 5.4%.
- **Point D** = inflation falling, CB begins easing.

Figure 3.4 The 3-equation model: the adjustment of the economy to a permanent demand shock.

The 3-Equation Model in Action: *Permanent Consumption Boom*



- **Long-Run Equilibrium: Point Z**

- With permanent boom, neutral real rate rises to 3%.
- Zero output gap condition $\rightarrow r = 3\%$.
- With $\pi = 2\%$: $i_Z = 5\%$.
- Output = potential, inflation = target.
- **Point Z** = new steady state: inflation back at 2%, output at equilibrium, but **policy rate permanently higher** (5% vs. 3% at A).

- **The Full Adjustment Path**

- **A $\rightarrow$ B:** Shock, inflation jumps to 3%, output +1%.
- **B $\rightarrow$ C:** CB reacts strongly, output gap shrinks, rate peaks.
- **C $\rightarrow$ D:** Expectations fall, PC shifts down, inflation easing.
- **D $\rightarrow$ Z:** Inflation returns to target, rates settle at 5% (higher than at A).

Figure 3.4 The 3-equation model: the adjustment of the economy to a permanent demand shock.

Modelling: deriving the MR Curve

Deriving the Monetary Rule (MR):

1. Define the central bank's preferences that capture its cost of deviating from inflation target and equilibrium output
2. Define the central bank's constraints from the supply side, i.e. the Phillips Curve (PC)
3. Derive the best response monetary rule in the output-inflation space, which gives the MR curve
4. Once the optimal output-inflation combination is determined using the MR, the central bank uses the IS curve to implement its choice (by setting the interest rate)

Modelling: the central bank's preferences

The central bank's preferences are given by a loss function:

$$L = (y_t - y_e)^2 + \beta(\pi_t - \pi^T)^2$$

- The higher the loss (L), the worse it is for the central bank
- The central bank is worse off the further inflation (π_t) is away from its target level (π^T), and the further output (y_t) is away from its equilibrium level (y_e)
- β reflects the relative degree of inflation aversion of the central bank

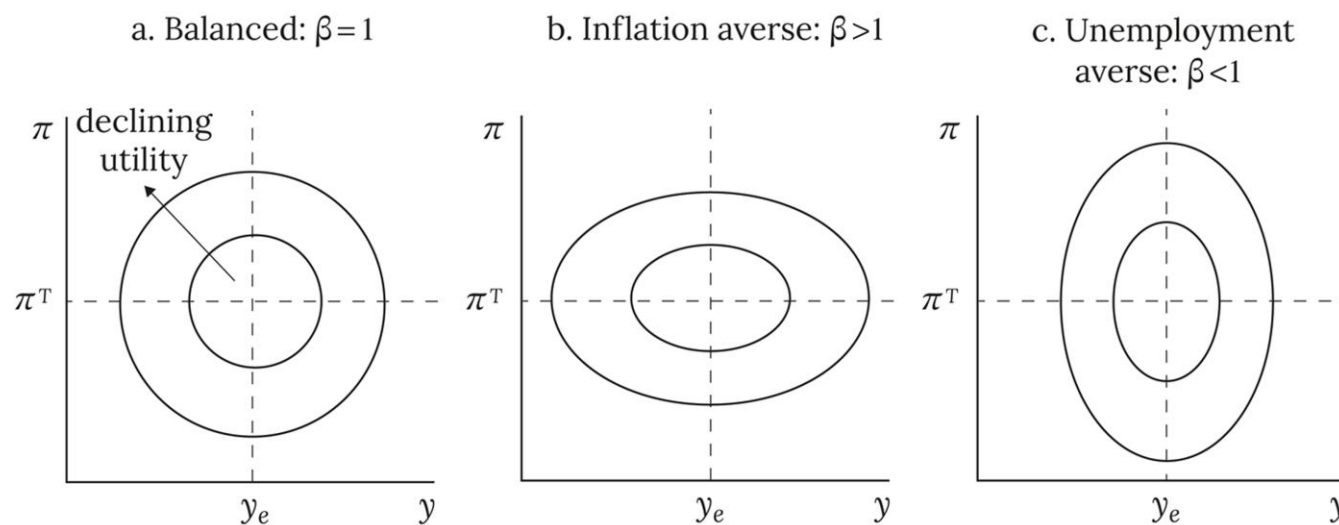


Figure 3.5 Central bank loss functions: utility declines with distance from the 'bliss point'.

Modelling: deriving the MR Curve

Deriving the Monetary Rule (MR):

1. Define the central bank's preferences that capture its cost of deviating from inflation target and equilibrium output
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Modelling: the Phillips curve constraint

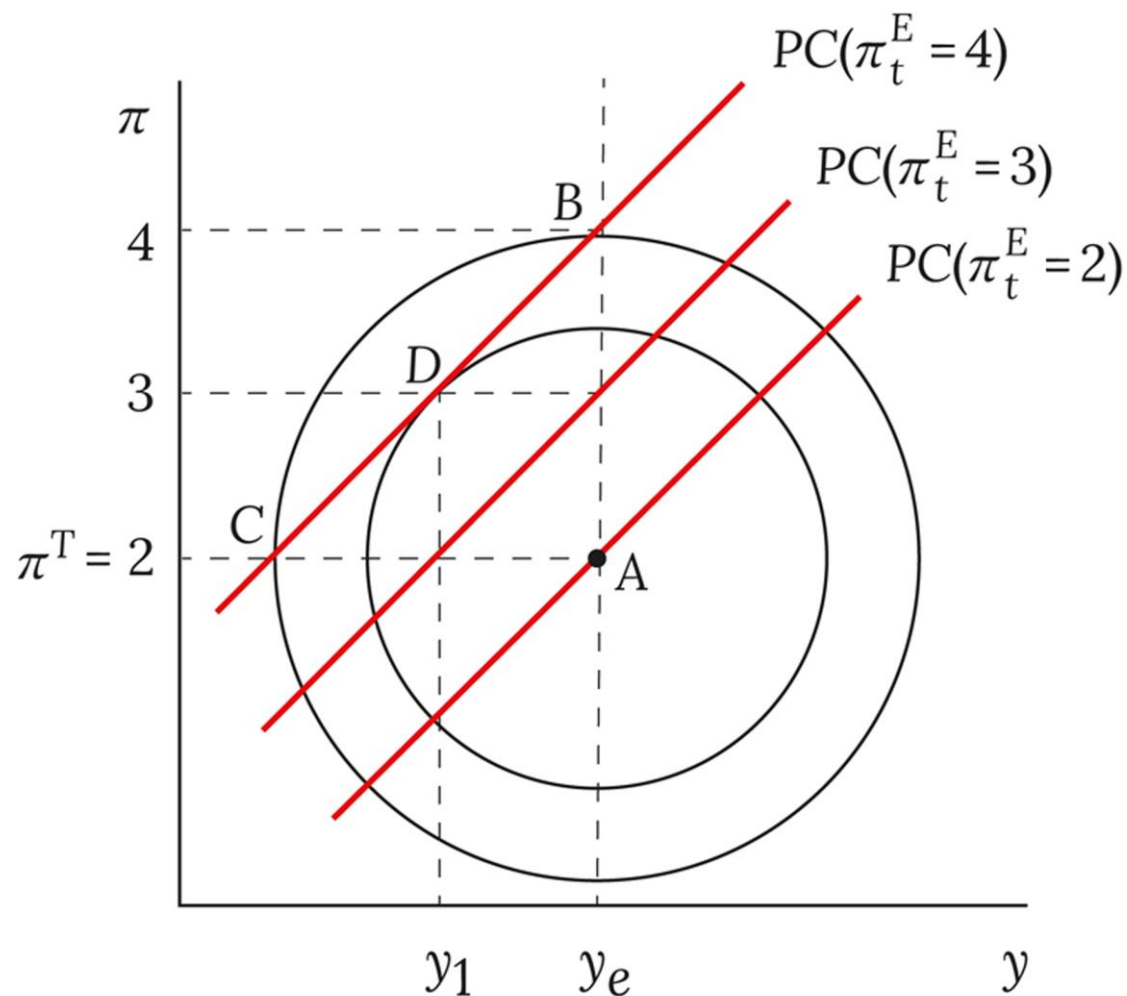


Figure 3.6 Loss circles and Phillips curves.

The Phillips curve is a constraint faced by the central bank:

$$\begin{aligned}\pi_t &= \pi_t^E + \alpha(y_t - y_e) \\ &= \pi_{t-1} + \alpha(y_t - y_e)\end{aligned}$$

The PC shows combinations of output and inflation which are achievable by the central bank, for a given inflation expectation

- If the economy is on $PC(\pi_t^E = 4)$, the 'bliss point' A at which loss is zero cannot be achieved
- The Central Bank faces a trade-off between output and inflation

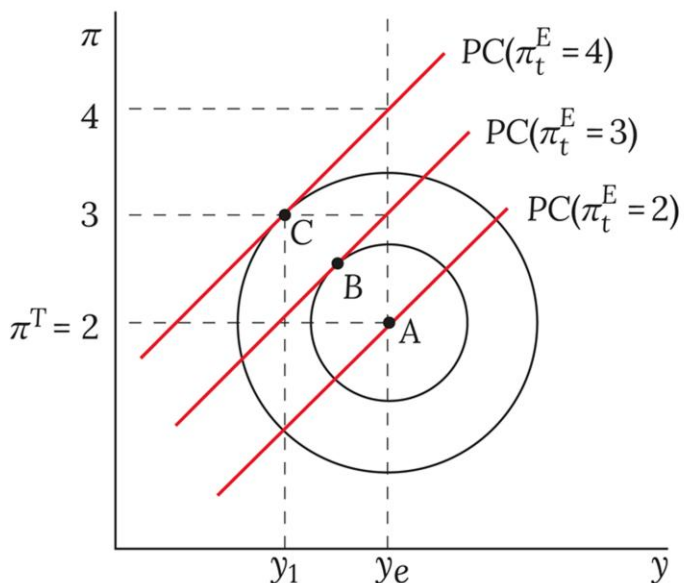
Modelling: deriving the MR Curve

Deriving the Monetary Rule (MR):

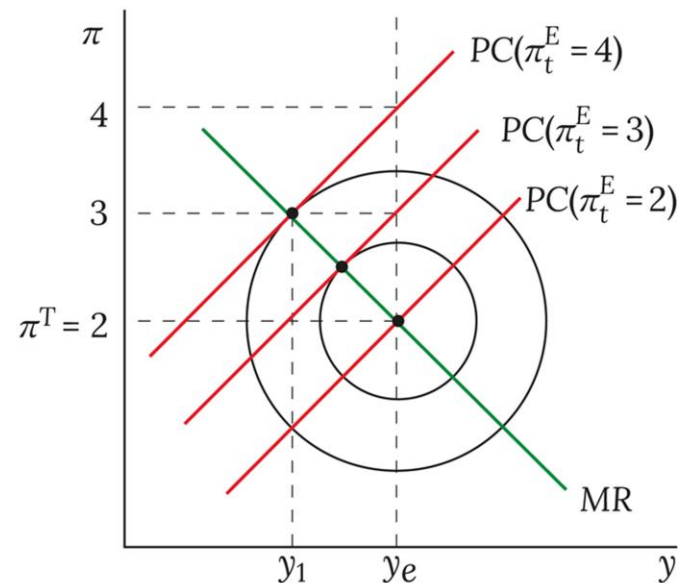
1. Define the central bank's preferences that capture its cost of deviating from inflation target and equilibrium output
2. Define the central bank's constraints from the supply side, i.e. the Phillips Curve (PC)
3. Derive the best response monetary rule in the output-inflation space, which gives the MR curve
4. Once the optimal output-inflation combination is determined using the MR, the central bank uses the IS curve to implement its choice (by setting the interest rate)

Modelling: graphical derivation of MR Curve

Step 1: Find the tangencies between the loss circles and the Phillips curves



Step 2: Join the tangencies to derive the MR curve



The MR shows the preferred output-inflation combination that minimizes the central bank's loss function for any PC curve it faces, i.e., tangency points between the PC and loss circles:

$$(y_t - y_e) = -\alpha\beta(\pi_t - \pi^T)$$

- It tells the CB the best-response output gap ($y_t - y_e$) to choose when inflation is away from target ($\pi_t - \pi^T \neq 0$)

Figure 3.7 Deriving the MR curve. Visit www.oup.com/he/carlin-soskice to explore deriving the MR curve with Animated Analytical Diagram 3.7.

Modelling: deriving the MR Curve

Deriving the Monetary Rule (MR):

1. Define the central bank's preferences that capture its cost of deviating from inflation target and equilibrium output
2. Define the central bank's constraints from the supply side, i.e. the Phillips Curve (PC)
3. Derive the best response monetary rule in the output-inflation space, which gives the MR curve
4. Once the optimal output-inflation combination is determined using the MR, the central bank uses the IS curve to implement its choice (by setting the interest rate)

Modelling: implementation of central bank policy

Now that the best response output gap ($y_t - y_e$) is chosen, it is implemented using the real interest rate (r) on the IS curve:

$$\begin{aligned} y &= k(c_0 + a_0 + G) - ka_1r \\ &= A - ar \end{aligned}$$

- In reality, the CB sets the nominal interest rate to achieve the desired real interest rate. This is called the Taylor Principle
- To take into account the delayed impact of interest rates on output, the 3-equation model uses the following dynamic IS curve:

$$y_t = A - ar_{t-1}$$

Modelling: implementation of central bank policy (cont.)

- **The Taylor Rule:** A simple guideline for how central banks set the policy interest rate in response to inflation and output.
- **Formula (basic version):**
- $i_t = r^* + \pi_t + \phi_\pi(\pi_t - \pi^*) + \phi_y(y_t - y^*)$
 - i_t : nominal policy interest rate
 - r^* : neutral real interest rate
 - π_t : current inflation
 - π^* : target inflation
 - $(y_t - y^*)$: output gap (actual – potential output)
 - ϕ_π, ϕ_y : response weights (how strongly the central bank reacts)
- **Interpretation:**
 - If inflation > target → raise rates.
 - If output > potential → raise rates.
 - If both below target → cut rates.
- **Why it matters:**
 - Provides a systematic rule for policy instead of guesswork.
 - Anchors expectations: households, firms, and markets can “predict” central bank moves.
 - Forms the foundation of the MR curve in the IS–PC–MR model.

Modelling: inflation shock

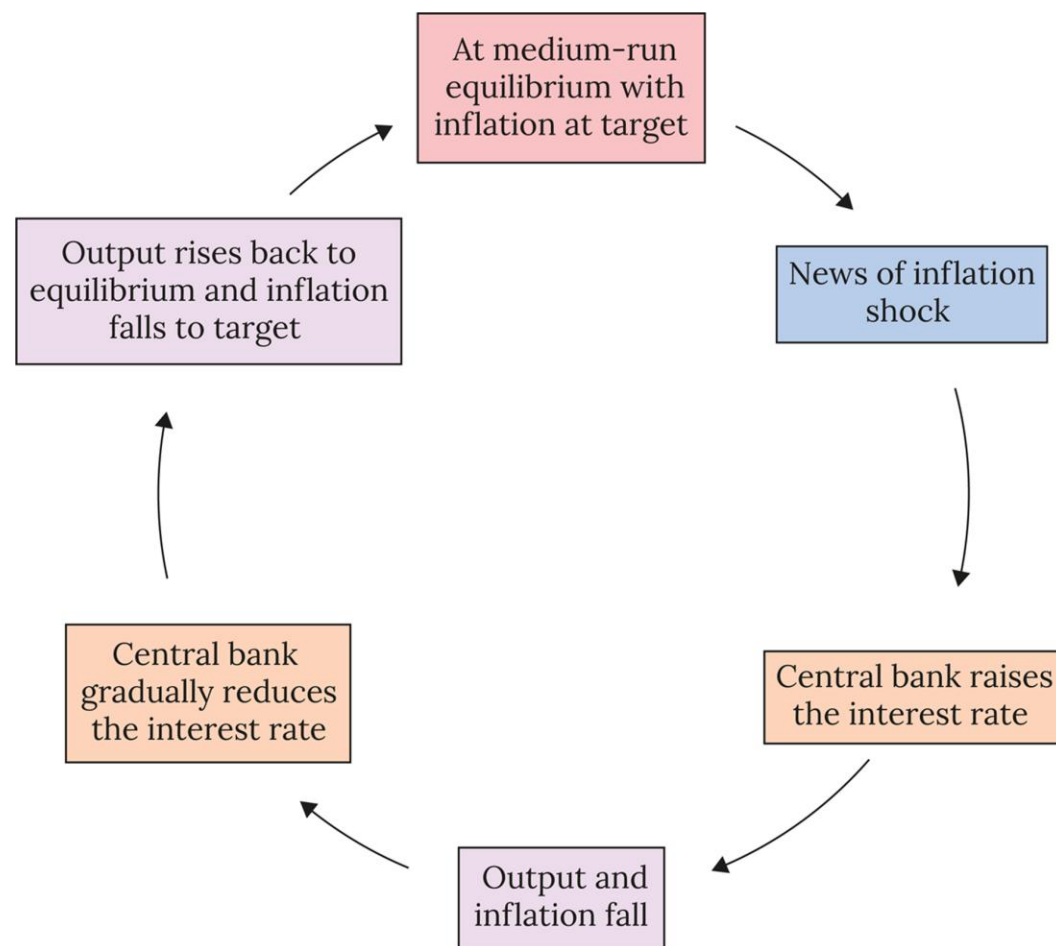


Figure 3.8 Dynamic adjustment to an inflation shock.

Inflation shock is represented as an exogenous shift in the Phillips Curve

- Causes of inflation shock: natural disaster, a shock to expected inflation, etc.

The central bank stabilize the economy after the inflation shock following two main steps:

1. Choose the position on the new PC that minimizes their loss
2. Find the optimal increase in the real interest rate with the IS curve to achieve the chosen position

a. 3-equation model

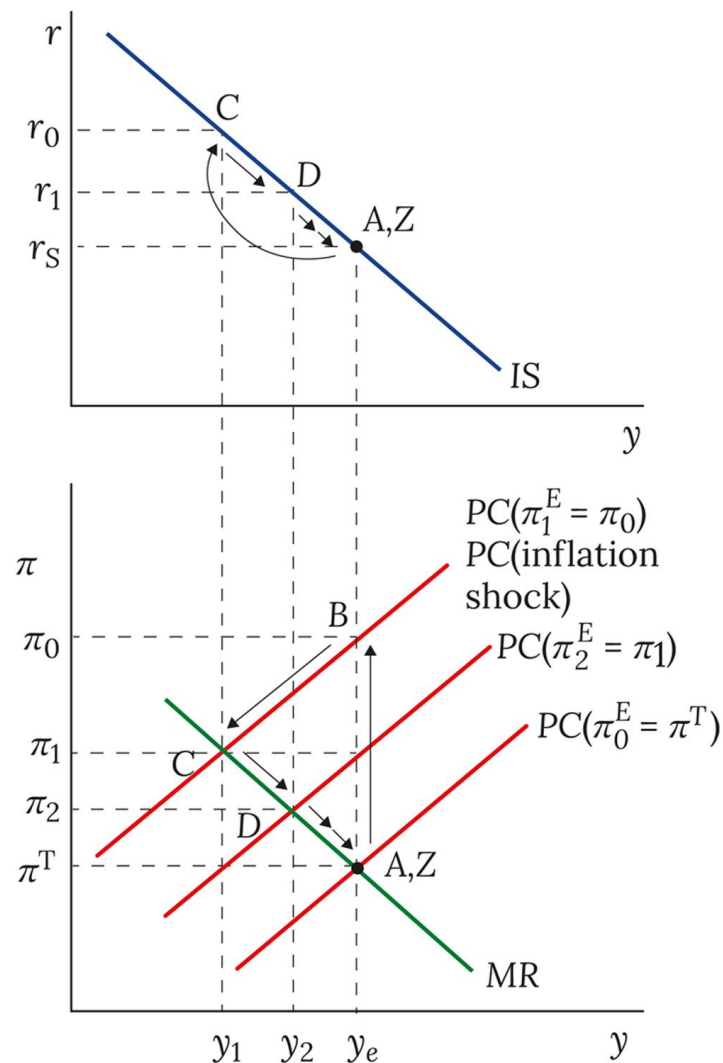


Figure 3.9(a) Inflation shock and the monetary rule. Visit www.oup.com/he/carlin-Soskice to explore an inflation shock and the monetary rule with Animated Analytical Diagram 3.9.

Modelling: inflation shock

Period 0:

- Economy is at the Bliss point A and then inflation shock hits
- PC shifts upwards to PC (inflation shock), economy moves to B
- The CB forecasts PC for the next period, $PC(\pi_1^E = \pi_0)$
- The CB chooses the optimal point C on this new PC which minimizes its loss and set interest rate at r_0 accordingly
- The economy ends with π_0 , y_e and r_0

Period 1:

- r_0 becomes effective: the economy moves to C with output at y_1 and inflation at π_1
- The CB forecasts PC for period 2 with the updated inflation expectation, $PC(\pi_2^E = \pi_1)$
- The CB chooses the optimal point D and set interest rate at r_1
- The economy ends with π_1 , y_1 and r_1

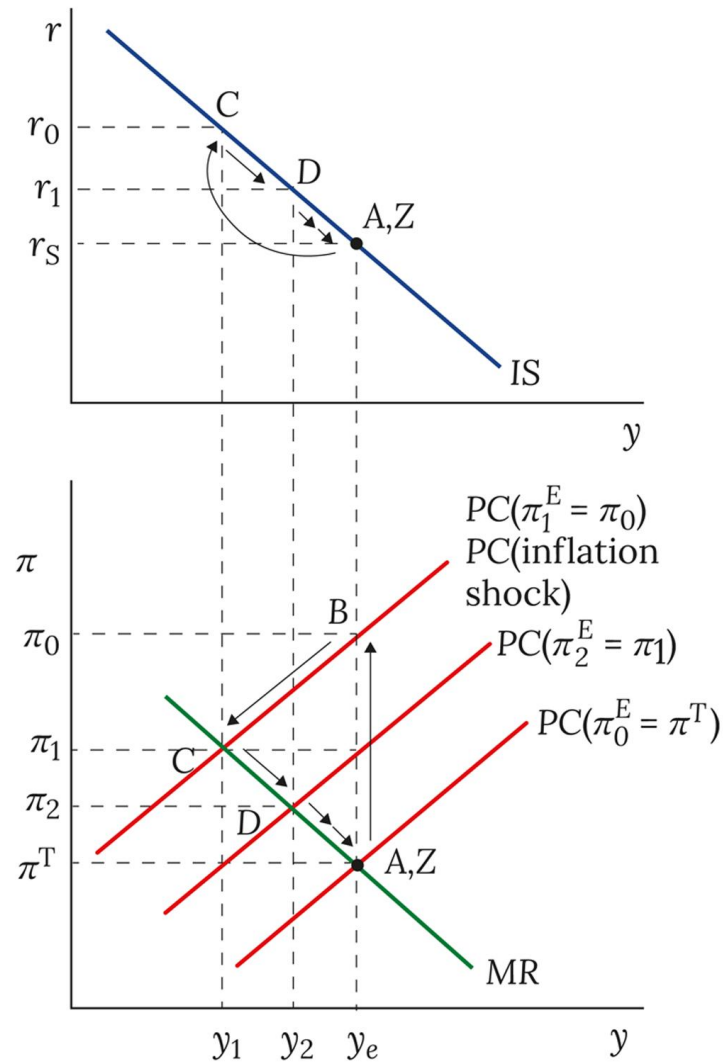
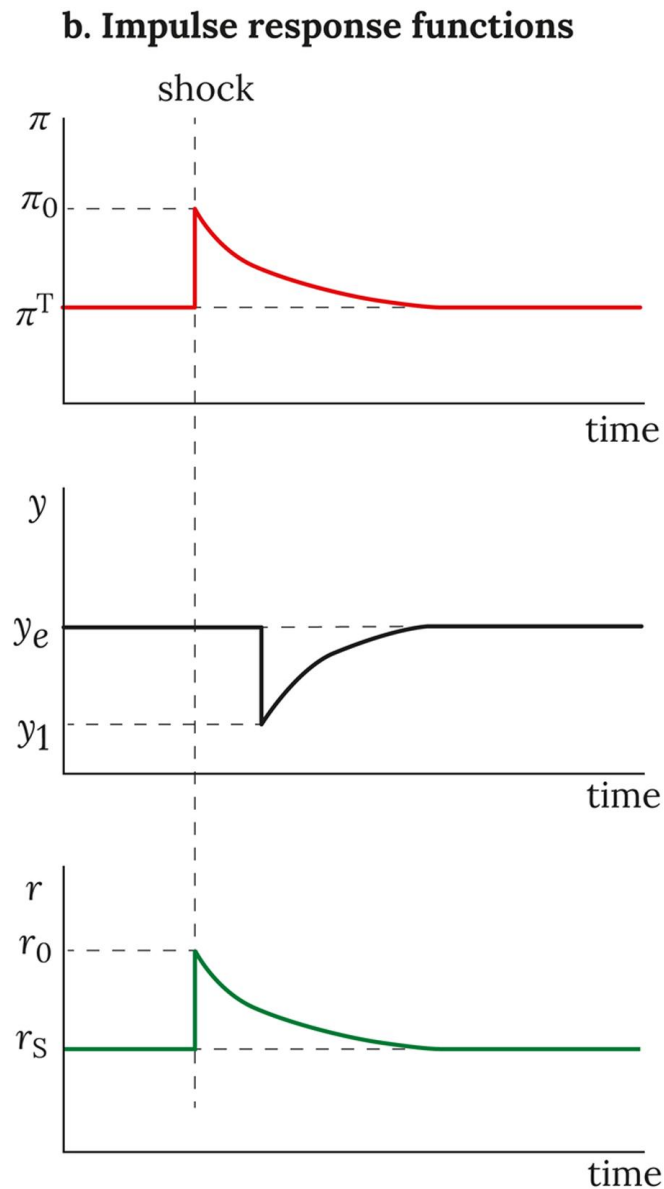


Figure 3.9(a) Inflation shock and the monetary rule. Visit www.oup.com/he/carlin-Soskice to explore an inflation shock and the monetary rule with Animated Analytical Diagram 3.9.

Period 2 onwards:

- Economy moves to D as the lower r_1 increases output to y_2 and inflation falls to π_2
- The same process repeats until the economy is back at its equilibrium Z
- The economy moves down along MR curve: The CB gradually adjusts r_1 down to r_S , output rises slowly from y_2 to y_e and inflation eventually falls from π_2 to π^T
- When the economy stabilizes, output is at y_e , inflation is at π^T and interest rate is at r_S

Modelling: impulse response function



The Impulse Response Function

- A function which shows the time path of macroeconomic variables following a shock
- Inflation π : increases after the shock and decreases gradually to π^T
- Output y : decreases in period 1 due to the 1-period lagged effect of r on y and increases gradually to y_e
- Interest rate r : increase after the shock and decrease gradually to r_S

Figure 3.9(b) Inflation shock and the monetary rule. Visit www.oup.com/he/carlin-Soskice to explore an inflation shock and the monetary rule with Animated Analytical Diagram 3.9.

a. 3-equation model

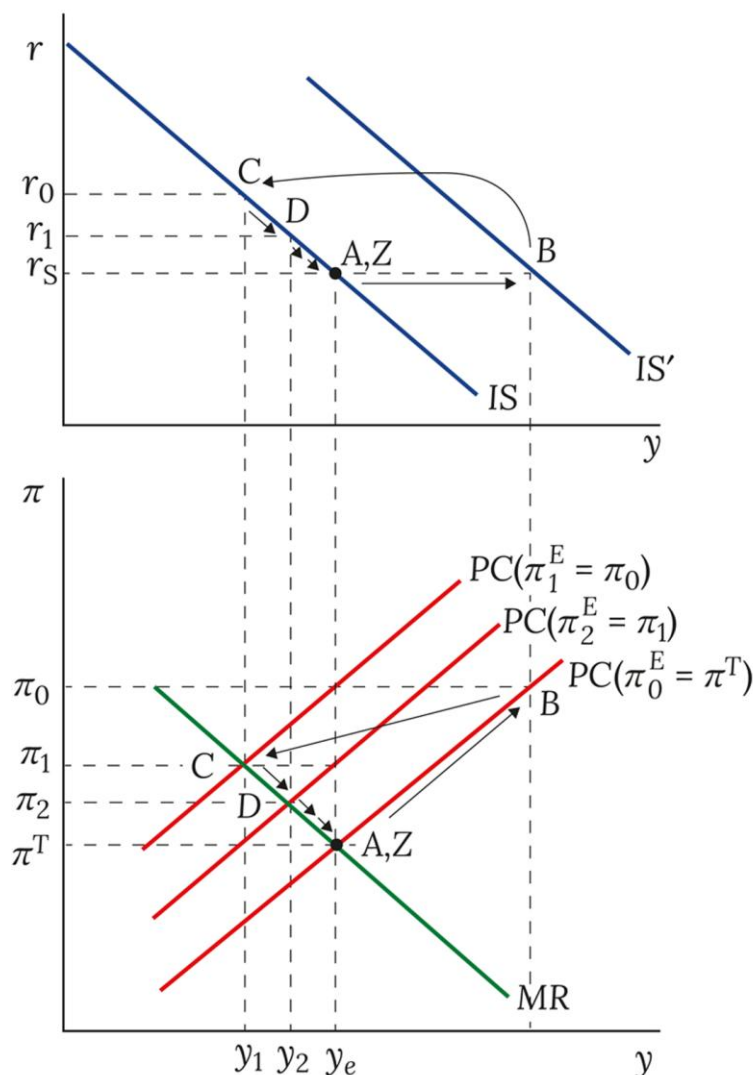


Figure 3.10(a) Adjustment of the economy to a temporary positive aggregate demand shock. Visit www.oup.com/he/carlin-soskice to explore the central bank's response to a temporary demand shock with Animated Analytical Diagram 3.10.

Application I: a temporary demand shock

Period 0:

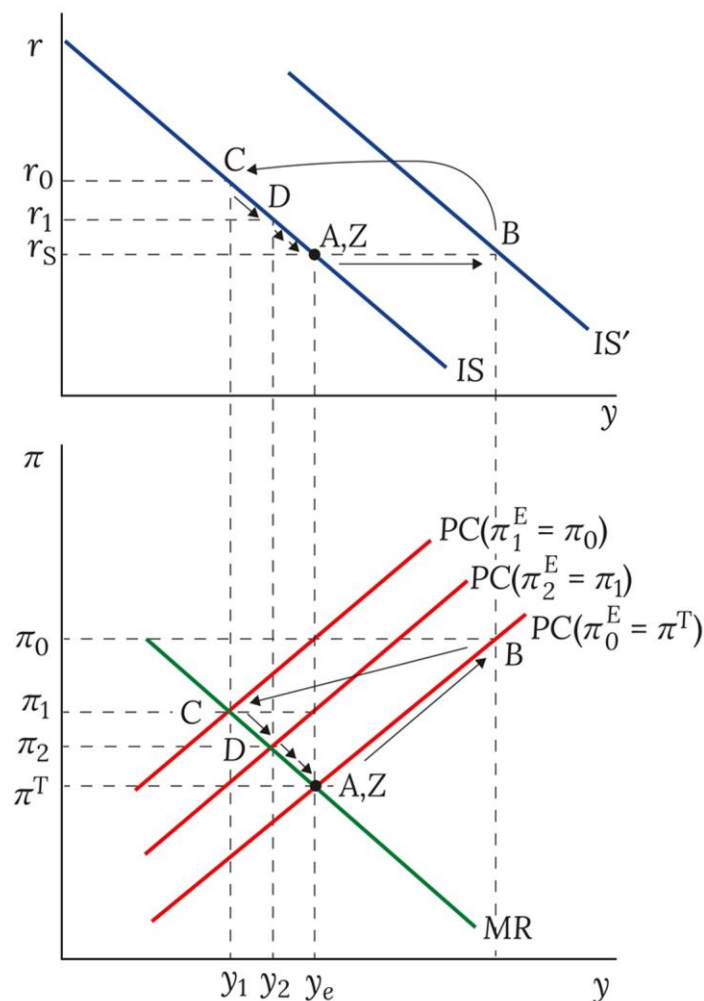
- Temporary AD shock: IS shifts to IS', but stays at IS' for **1 period only**
- The economy shifts from initial Bliss point A to B
- The CB forecasts the PC for period 1, which is $PC(\pi_1^E = \pi_0)$
- The CB expects that IS' returns to IS in period 1, and sets r_0 with IS instead of IS'
- The economy ends with π_0, y_0 and r_0

Period 1:

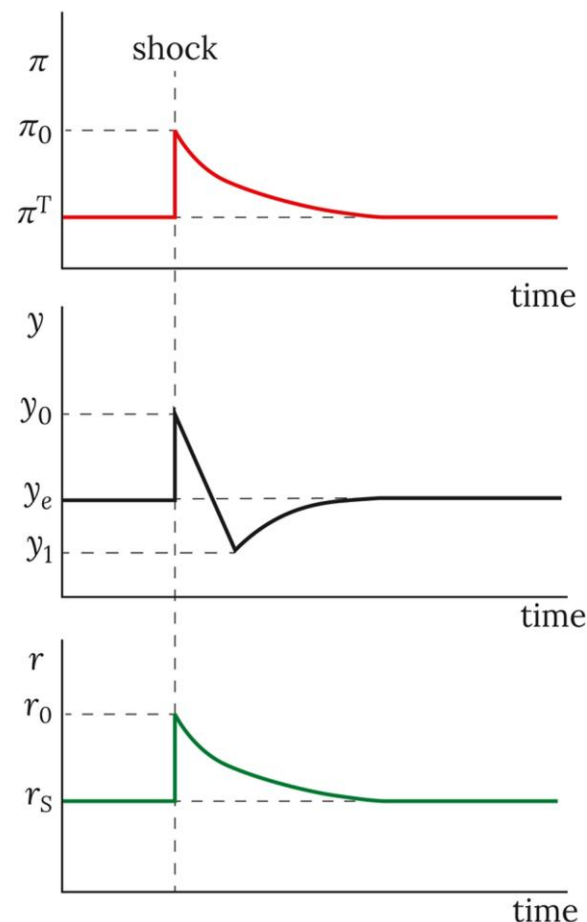
- The economy moves to C with output at y_1 and inflation at π_1
- The CB forecasts PC for period 2 to be $PC(\pi_2^E = \pi_1)$ and sets interest rate r_1 to achieve D
- The economy ends with π_1, y_1 and r_1

Application I: a temporary demand shock

a. 3-equation model



b. Impulse response functions



Period 2 onwards:

- The economy moves to D as demand increases from lower r_1
- The same process repeats and r is gradually reduced until the economy reaches its equilibrium Z with π^T , y_e and r_s

Figure 3.10 Adjustment of the economy to a temporary positive aggregate demand shock. Visit www.oup.com/he/carlin-soskice to explore the central bank's response to a temporary demand shock with Animated Analytical Diagram 3.10.

Application: forecasting and lags

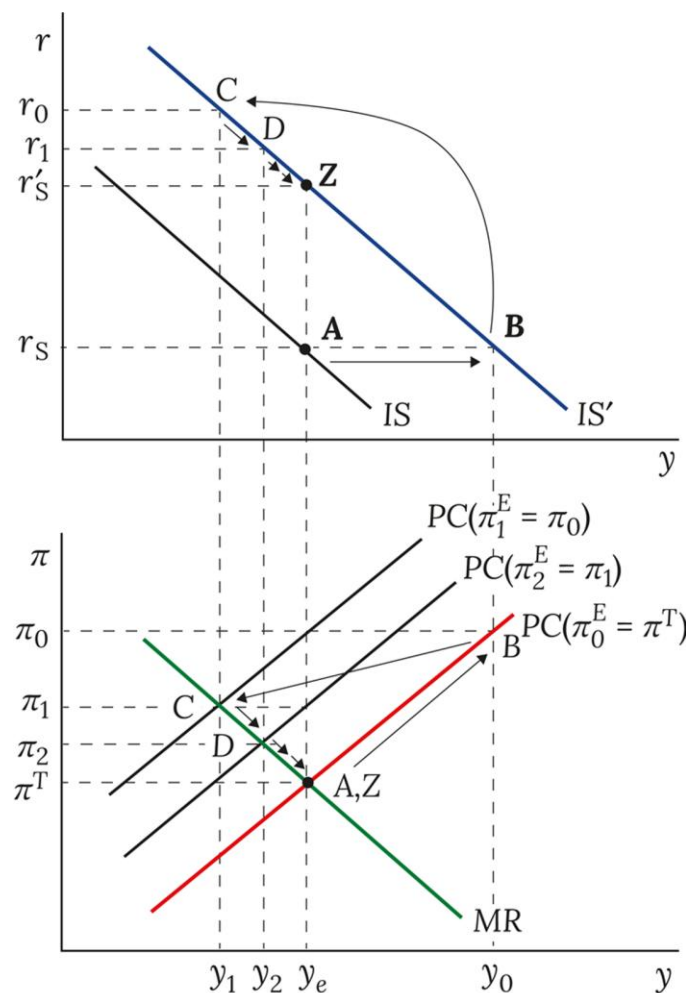


Figure 3.11 Adjustment of the economy to a permanent positive aggregate demand shock. Visit www.oup.com/he/carlin-soskice to explore the central bank's response to a permanent demand shock with Animated Analytical Diagram 3.11.

To respond to an aggregate demand shock, CB has to forecast both the PC and IS curve

- Forecasting the IS curve involves predicting the length of the shock: temporary or permanent
- The persistence of shocks affects the CB's optimal response
 - Permanent shock: IS' does not shift back to IS
 - A larger initial increase in r (i.e., from r_s to r_0) is needed compared to a temporary shock
 - The new equilibrium r'_s is higher

Application: forecasting and lags

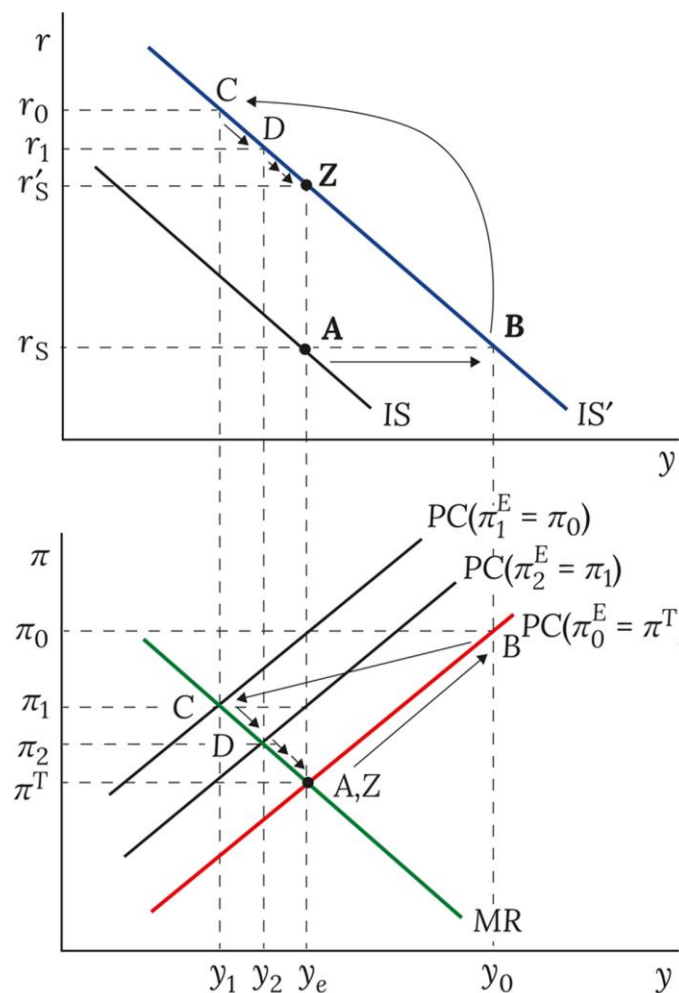


Figure 3.11 Adjustment of the economy to a permanent positive aggregate demand shock. Visit www.oup.com/he/carlin-soskice to explore the central bank's response to a permanent demand shock with Animated Analytical Diagram 3.11.

In the benchmark 3-equation model, we have assumed a one period lag. If there is no lags and interest rate can affect output immediately, we have:

$$y_t = A - ar_t$$

In this case of a permanent positive AD shock, CB can raise the interest rate to r'_S right after the shock

- The economy goes directly from A to Z in the IS diagram and it remains at A in the PC-MR diagram
- Inflation stays at target as the shock is fully offset

The central bank has a greater chance to stabilize the economy and inflation if their forecast is timely and accurate

Application II: deflation trap

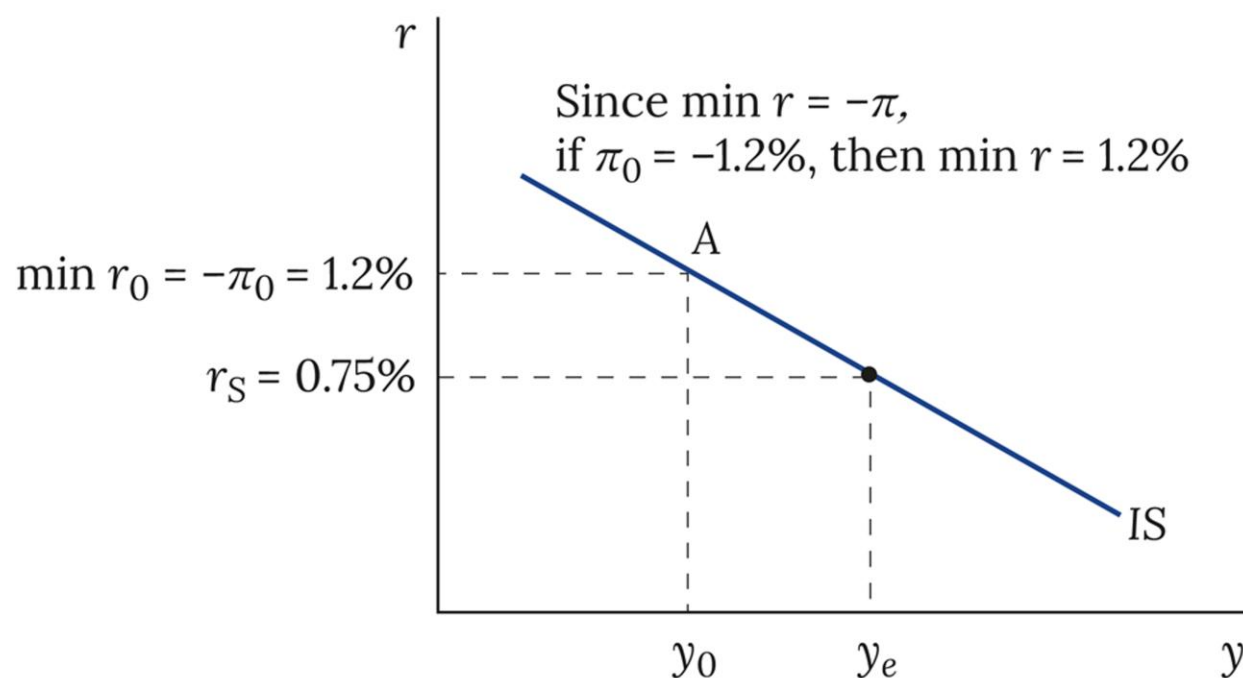


Figure 3.12 The zero lower bound on the nominal interest rate.

Recall the Fisher equation from Ch. 1: $i = r + \pi^E$

- The Zero Lower Bound (ZLB) on i states that $\min i = 0$, which implies that $\min r = -\pi^E$
- If expected inflation is -1.2% , the minimum r attainable is 1.2% which is above r_S
- r cannot be reduced below 1.2% to achieve y_e , the ZLB is hit, and the economy is stuck at a depressed level of output y_0

Application II: deflation trap

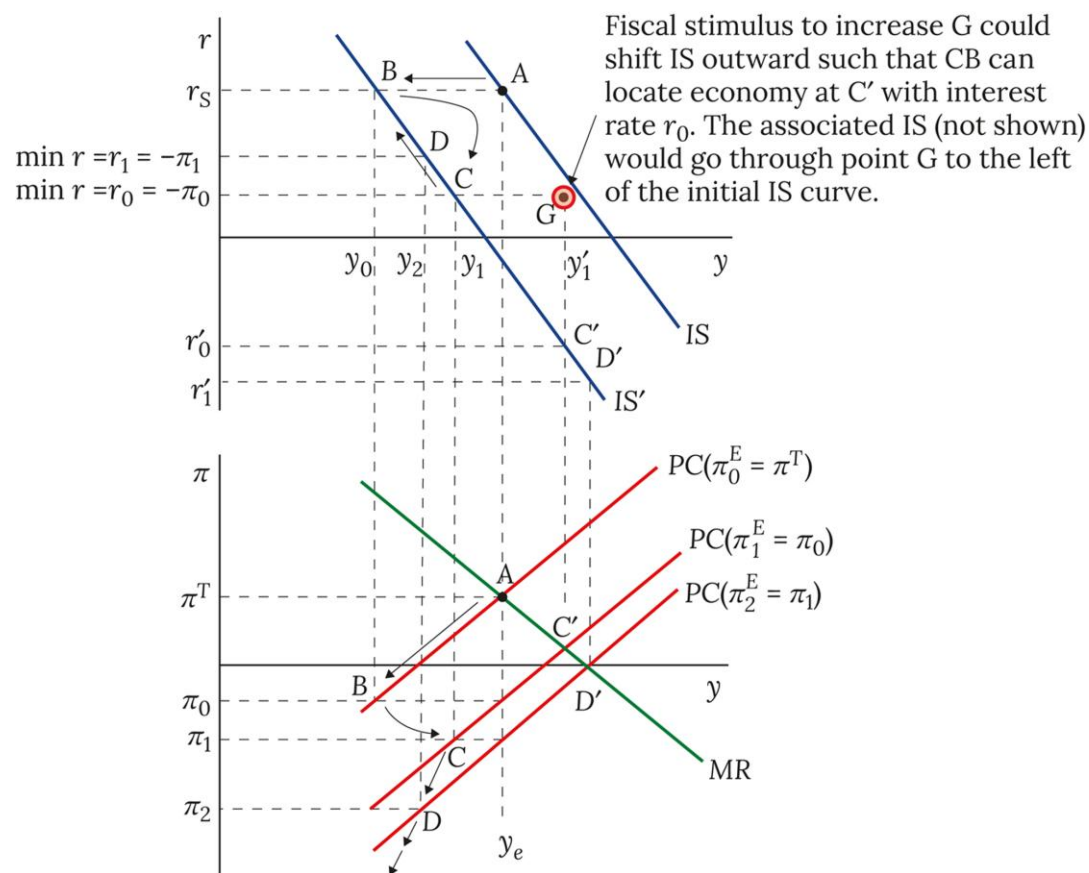


Figure 3.13 How a large negative permanent aggregate demand shock can lead to the economy entering a deflation trap. Visit www.oup.com/he/carlin-soskice to explore a deflation trap with Animated Analytical Diagram 3.13.

The deflation trap with the 3-equation model

Period 0:

- Large permanent negative AD shock: the IS shifts to IS'
- Inflation is now negative (deflation), as the economy moves to point B
- Given the forecasted $PC(\pi_1^E = \pi_0)$, the CB would like to choose C' on the MR, which requires setting r'_0
- However, r'_0 is below the minimum that can be achieved by setting a zero nominal interest rate, so the CB can only set the minimum $r_0 = -\pi_0$
- The economy ends with π_0 , y_0 and r_0

Application II: deflation trap

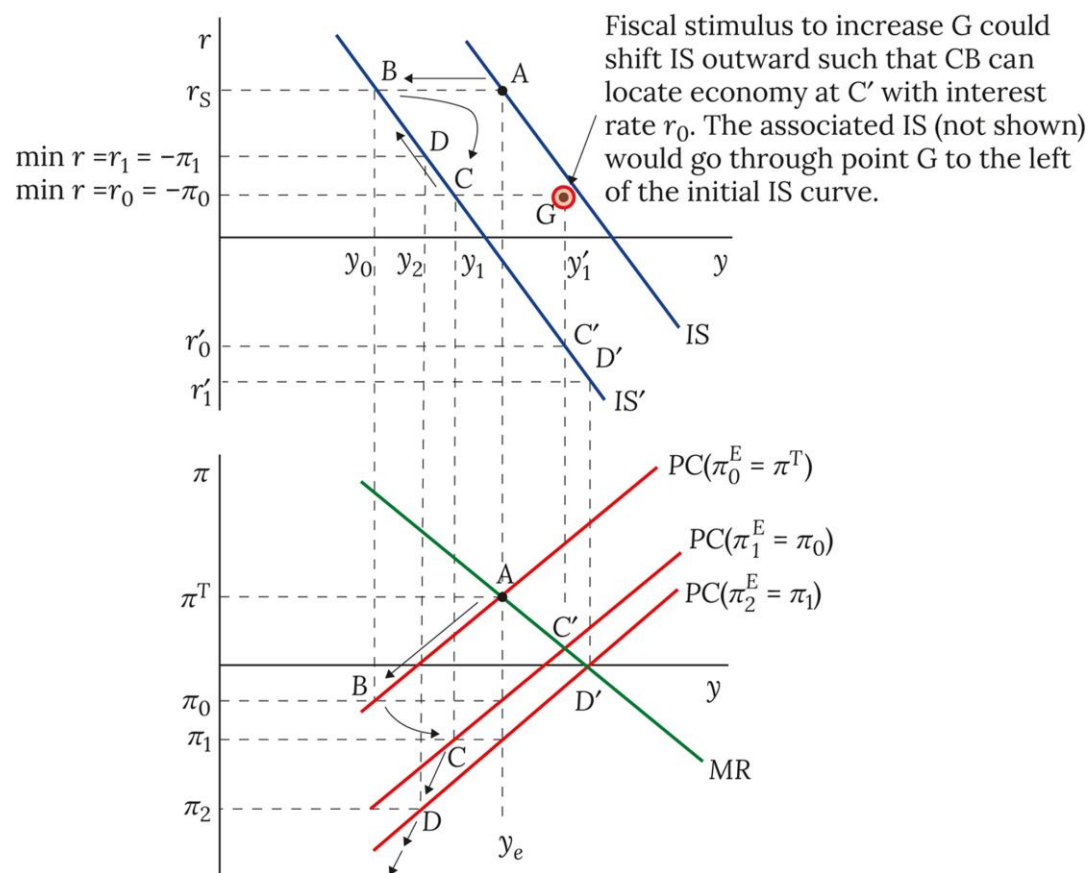


Figure 3.13 How a large negative permanent aggregate demand shock can lead to the economy entering a deflation trap. Visit www.oup.com/he/carlin-soskice to explore a deflation trap with Animated Analytical Diagram 3.13.

Period 1:

- The economy moves to C
- Inflation falls further to π_1 and the CB forecasts the next period's PC $PC(\pi_2^E = \pi_1)$ and chooses D' , which requires setting r_1'
- However, the minimum attainable real interest rate is $r_1 = -\pi_1$
- Thus, the economy ends this period with π_1 , y_1 , and r_1

Application II: deflation trap

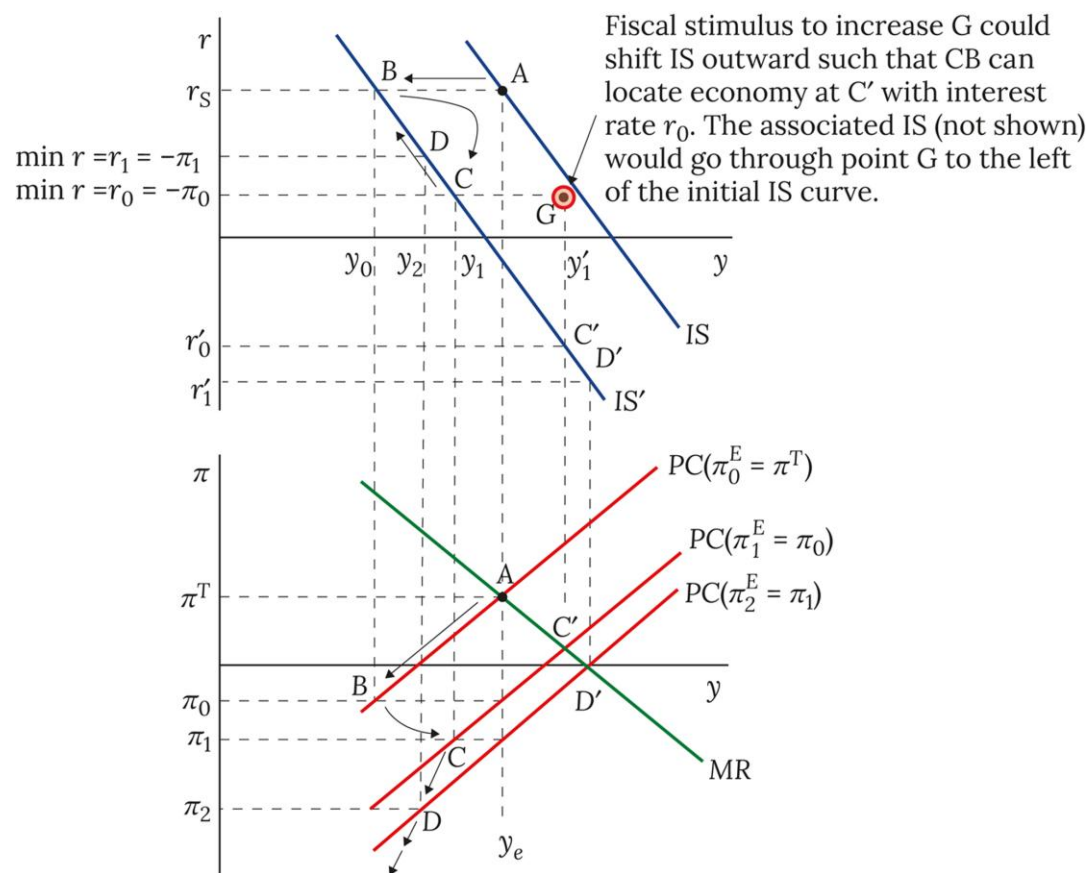


Figure 3.13 How a large negative permanent aggregate demand shock can lead to the economy entering a deflation trap. Visit www.oup.com/he/carlin-soskice to explore a deflation trap with Animated Analytical Diagram 3.13.

Period 2 onwards:

- The economy moves to D: the higher interest rate r_1 dampens demand, output and inflation both fall to y_2 and π_2 respectively
- The economy enters a **deflation trap**: in each future period, inflation falls further, causing r to increase and output to fall continuously
- Conventional monetary policy is ineffective in reverting the economy back to its medium-run equilibrium

Application II: deflation trap

There are policies that can escape the deflation trap

- Shifting the IS rightwards (to go through point G) after the initial AD shock
 - This can result from a spontaneous recovery of autonomous investment, a recovery of autonomous consumption or implementation of fiscal policy
 - CB can then achieve desired point C' by setting interest rate
- Creating more positive inflation expectations
 - If inflation expectations are less negative (π^E higher), then $\min r$ is lower and the PC curve shifts up
 - conventional MP now works to bring the economy back to equilibrium
- Unconventional Monetary Policy (Ch. 5 and 6)

Application III: supply shock

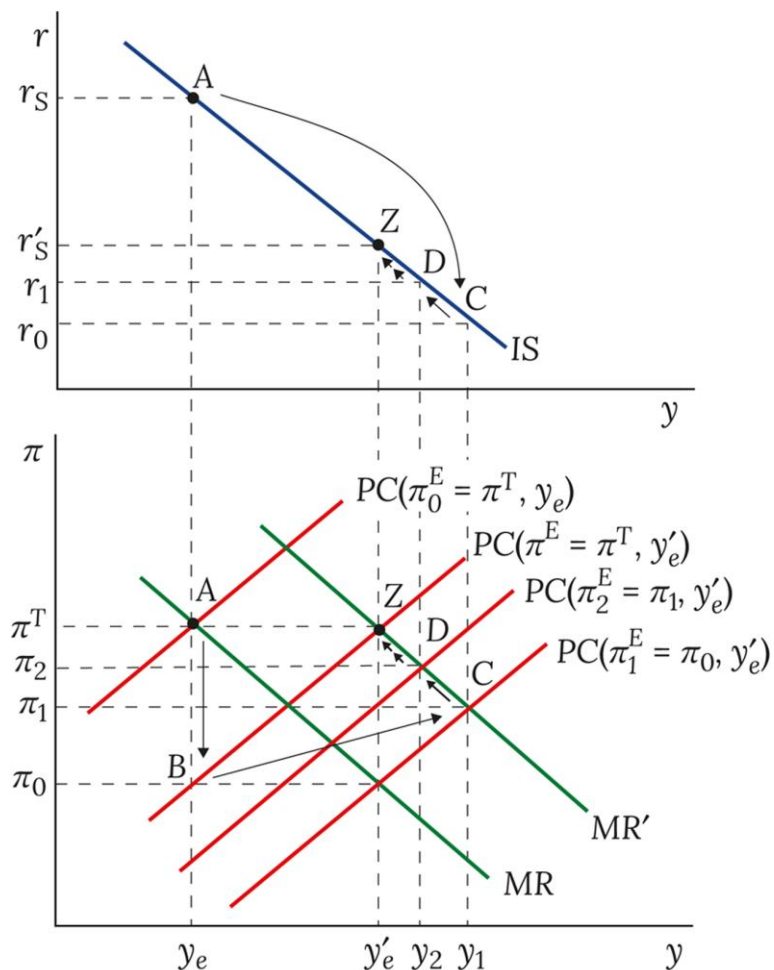


Figure 3.14 The adjustment of the economy to a positive permanent aggregate supply shock. Visit www.oup.com/he/carlin-soskice to explore a positive supply shock with Animated Analytical Diagram 3.14.

Supply shocks differ from demand and inflation shocks in 3 key ways:

- The equilibrium output changes
- MR curve shifts as it needs to go through the new equilibrium output
- Phillips curves need to go through the new equilibrium output

Application III: supply shock

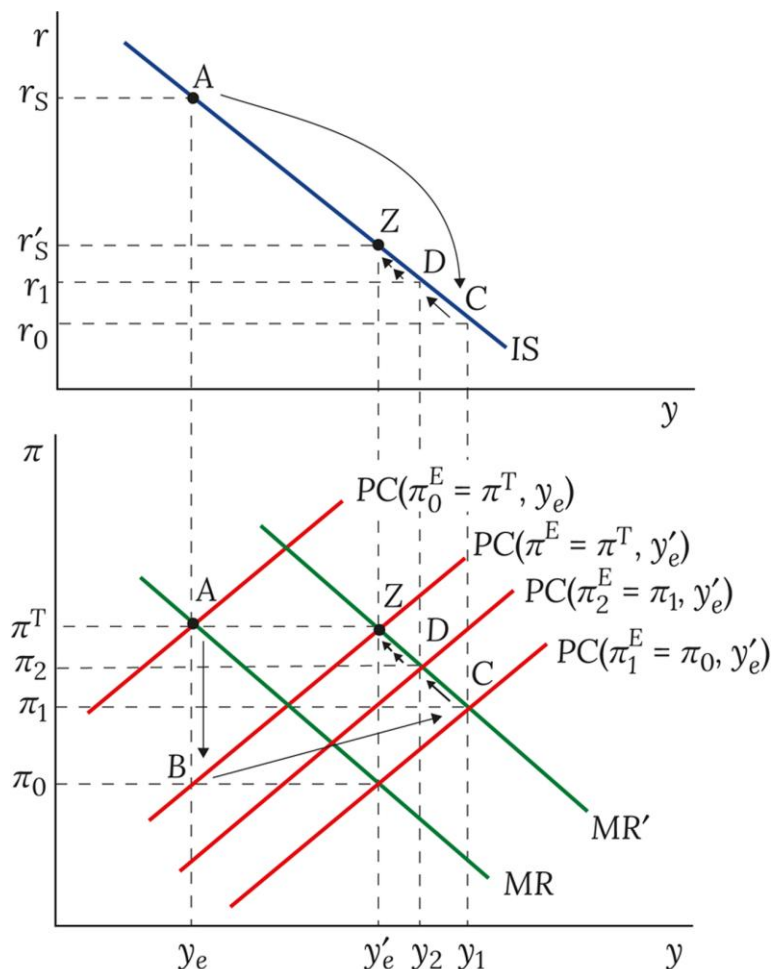


Figure 3.14 The adjustment of the economy to a positive permanent aggregate supply shock. Visit www.oup.com/he/carlin-soskice to explore a positive supply shock with Animated Analytical Diagram 3.14.

Period 0:

- Permanent supply shock: equilibrium output increases from y_e to y'_e
- Phillips curve shifts down from $PC(\pi_0^E = \pi^T, y_e)$ to $PC(\pi_0^E = \pi^T, y'_e)$
- The economy moves from A to B. Output is still at y_e , yet inflation has fallen to π_0
- The CB shifts MR to MR' and their bliss point is now Z that go through π^T and y'_e
- The CB forecasts $PC(\pi_1^E = \pi_0, y'_e)$ and set interest at r_0 to achieve C on MR'
- The economy ends with π_0, y_e and r_0

Application III: supply shock

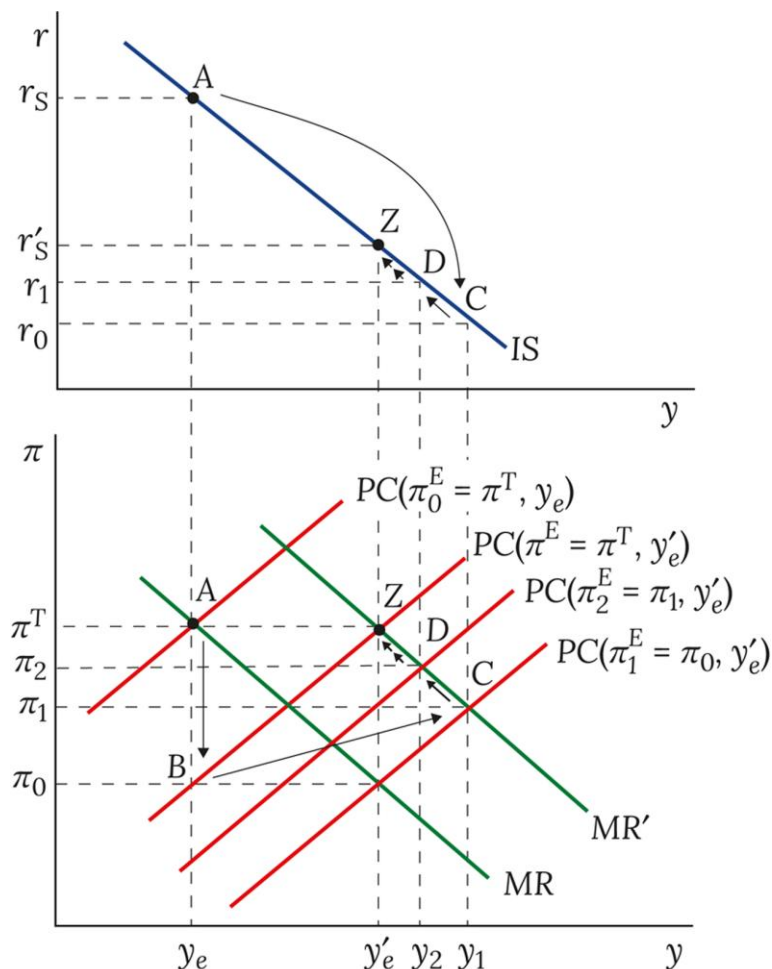


Figure 3.14 The adjustment of the economy to a positive permanent aggregate supply shock. Visit www.oup.com/he/carlin-soskice to explore a positive supply shock with Animated Analytical Diagram 3.14.

Period 1:

- The economy moves to C with output at y_1 and inflation at π_1
- The CB forecasts PC for period 2 to be $PC(\pi_2^E = \pi_1, y'_e)$ and sets interest rate r_1 to achieve D on MR'
- The economy ends with π_1, y_1 and r_1

Period 2 onwards:

- The economy moves to D
- The same process repeats until the economy reaches its new equilibrium Z with π^T, y'_e and r'_s

Summary

The equations of the 3-equation model are:

IS Curve: $y_t = A - ar_{t-1}$

PC Curve: $\pi_t = \pi_{t-1} + \alpha(y_t - y_e)$

MR Curve: $(y_t - y_e) = -\alpha\beta(\pi_t - \pi^T)$

- The IS curve models the demand side, the PC curve models the supply side and the MR curve models the policy maker

Central banks target low and stable inflation to minimize the negative effects of inflation on the economy and to avoid high and rising inflation

- Dangers of the deflation trap also discourage policy makers from targeting a 0% inflation rate

Summary

Central Banks aim to achieve their inflation target:

- The Monetary Rule (MR) shows the output and inflation combinations which minimize the CB's loss function
- The PC forms a constraint to the CB's choice of output gap
- The desired output gap is implemented by the CB using the real interest rate r . This is done following the IS curve
- Demand shocks cause a deviation of output from equilibrium while supply shocks move the economy to a new equilibrium

Role of lags and forecasting in monetary policy:

- The 3-equation model reflects that it takes time for the real interest rate to affect output
- Hence, forecasting (e.g., PC) is vital for the process of setting interest rates